Looking ahead? The Role of Future Educational Opportunities on Pre-College Human Capital

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Abstract

How do future schooling opportunities shape early educational decisions? In this paper, I exploit a large expansion in the supply of free higher education institutions (HEIs) in Brazil to analyze the role of tertiary education opportunities in increasing returns to basic schooling in low- and middle-income countries (LMICs). To understand how higher education expansion can increase demand for pre-tertiary schooling, I develop a simple framework that highlights the channels of impact and the necessary conditions for downstream effects to occur. Empirically, I employ a novel dataset on HEI roll-out combined with rich administrative data, and estimate the effect of receiving an HEI on local public students' pre-college outcomes using a staggered difference-in-differences design. Students in treated municipalities are 2% more likely to graduate from high school and 20% more likely to take a college-entry exam. These effects are driven solely by increased access to higher education rather than by local labor market changes, with local students' enrollment in federal HEIs increasing by 70% following the policy. Consistent with the framework, empirical evidence supports the prediction that the college premium is key to generating downstream effects: changes in high school graduation are positive only in municipalities with a high pre-policy college premium, and largest where the college premium is high and the secondary premium is low. Expanding access to higher education in contexts where college returns are high can therefore be a powerful tool for improving early educational attainment, especially in LMIC settings where stand-alone returns to high school are not enough to keep students in school.

Keywords: Returns to education, higher education, college premium, educational attainment

JEL Classification: O12, O15, O54, I25, I26

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1 Introduction

Barriers to education in low- and middle-income countries (LMICs) have declined substantially in recent decades, yet a significant share of students remains out of school. By 2023, over half of all countries included free primary and secondary education in their legal frameworks (UNESCO, 2025). In Brazil, the context of this paper, primary education has been free and mandatory for more than 50 years, while free high school was formally guaranteed as a right in 1988 and became compulsory in 2009. The number of public secondary schools more than tripled between 1990 and 2009. Yet in 2019, only 77.3% of 15- to 17-year-olds attended high school (Todos Pela Educação, 2025). Although these numbers reflect both supply and demand factors, completion remains low even among students who access secondary education. In the late 2000s, fewer than 60% of public school students who started high school graduated within five years.

Low returns to schooling do not appear to explain this lack of demand for schooling in Brazil. In the 2000s, despite substantial premiums of 39% for completing primary school and an additional 33% for completing high school, adults attained, on average, fewer than six years of schooling, and more than 10% of 10- to 14-year-olds were economically active (Edmonds and Pavcnik, 2005). These patterns suggest that individuals face high opportunity costs that existing returns to primary and secondary education do not offset. At the same time, returns to education are highly convex, especially in LMI contexts where tertiary returns are 2 to 3 times as large as the secondary premium, as measured by Gethin (2025)[1]. These high, often unreachable returns at the top of the schooling distribution might provide an opportunity to pull constrained students up the educational ladder. In this paper, I ask: what happens when higher education returns become more accessible?

I explore an ambitious federal program in Brazil from 2003 to 2014 that more than doubled the supply of public and free four-year colleges and two-year trade schools. I evaluate whether the arrival of a new higher education institution (HEI) affects local public students' pre-tertiary persistence, measured by the likelihood of starting and graduating high school, and academic performance, measured by standardized test scores in grade 9 and the college-entry exam.[2]

To understand how and when higher education expansion can influence pre-tertiary educational decisions, I develop a simple framework in which a forward-looking 9th-grade student must choose whether to continue to high school and to complete post-secondary schooling after graduating. I

[1] In Brazil in the 2000s, completing higher education was associated with a 300% increase in earnings compared to completing secondary education.

[2] The policy includes expanding both universities and trade schools, which traditionally offer technical and professional courses, and majors focused on technical skills, although some do offer traditional majors.

identify which necessary conditions need to be in place for students to change their pre-tertiary schooling decision after higher education opportunities are introduced. I also identify two channels through which a new HEI might shift the share of students completing high school: a direct educational channel, by increasing the probability of entering tertiary education and directly decreasing tertiary education attendance costs, and an indirect labor market channel, by changing local wage levels. Most importantly, this framework highlights the key role of the college premium in driving high school graduation by increasing the expected returns to high school, especially in places where returns to high school are modest.

Empirically, I estimate the impact of higher education expansion on pre-tertiary students' outcomes using a staggered difference-in-differences approach proposed by Callaway and Sant'Anna (2021). I build a novel dataset of HEI openings using information from institutional documents, HEIs' websites, legal ordinances, and data from the Brazilian Ministry of Education, combined with rich administrative data from the Ministry of Education that contains the universe of K-12 enrollment and of individuals who take the national college-entry exam. I find that students in municipalities that received an HEI as part of this program were 2% more likely to eventually graduate from high school, but no more or less likely to graduate on time. Students are also 20% more likely to take a national college-entry exam, with no decrease in scores. I find no significant impacts on performance, measured either at 9th grade or at the end of high school.

This policy provides a unique setting where different municipalities received different types of HEIs. Comparing policy effects by type of HEI received can inform on whether different types of post-secondary education affect pre-tertiary students' decisions in different ways. I find that, while both universities and trade schools increase high school completion, only universities change the decision to enroll in high school in the first place. This is evidence that the marginal students moved by trade schools and universities are not the same. Trade school opportunities seem more salient to students who would have transitioned to high school but dropped out in the absence of the policy, but now choose to stay and graduate. Universities change a student's decision to start high school in the first place.

Throughout, all of my analysis focuses on public school students who are more likely to come from low-income households. However, within public schools, non-white students are especially disadvantaged on all metrics of pre-tertiary persistence and performance. I further analyze heterogeneity by race to identify who is most likely to benefit from the policy. I find that only non-white students are responding to new higher education opportunities by changing their decision to start high school and, conditional on doing so, experience a 4% increase in the likelihood of graduating high school, nearly four times the effect observed for white students (although only marginally statistically different). Non-white students are also driving the observed increase in the share of students choosing to take a college-entry exam.

The 2% increase in high school graduation following this higher education policy represents 1/8 to 1/6 of the effects found in quasi-experimental studies measuring the impact of being admitted to secondary schooling or receiving a high school voucher (Ozier, 2018; Angrist et al., 2006). Although the effect is seemingly small in magnitude, this policy did not change the supply of high school opportunities or directly target high school completion. I use the theoretical framework's predictions to conduct empirical exercises that provide suggestive evidence on the key channels of impact that might be mediating the observed effect.

While the framework identifies two channels of impact, I find that the observed effects are likely driven by changes in educational opportunities rather than changes in the local labor market. Local public students experience a nearly 1-percentage-point increase in enrollment following the opening of an HEI, representing a 70% increase relative to the pre-policy mean. On the other hand, I find no significant changes in formal-sector wages that could account for the observed positive effects. As predicted by the framework, changes in high school completion following an expansion in higher education opportunities should be most prevalent in contexts where the college premium is high and the high school premium is modest. I empirically assess this claim and find that the effects are positive only in municipalities with a high college premium pre-policy, and largest in municipalities with an above-median college premium and below-median high school returns.

Last, to assess whether higher education opportunities might be a sustainable channel to increase returns to high school, I examine how college and high school premiums in the formal labor market change over the next 6 years after the opening of an HEI. I find that college premiums remain stable, but high school premiums decrease sharply over time. This drop in high school premiums is driven by an increase in primary-education-level wages, which might be a direct result of a decrease in the supply of unskilled labor due to changes in educational attainment stemming from this policy, a result similar to that found by Khanna (2023) for primary education expansion in India. Provided that the college premium remains sizeable (a reasonable assumption given the limited supply of higher education), higher education opportunities should continue to shape pre-tertiary schooling decisions in the long run.

There are three key challenges for my identification strategy. First, treated municipalities were not randomly selected by the federal government and could be on a different trend than control municipalities. To bypass this selection problem, I control for a number of municipality-level characteristics observed by the government when choosing receiving municipalities, as stated in institutional documents, and relevant characteristics documented in the literature, such as political alignment (Arroyo and Petterini, 2020). Second, to check whether treated municipalities were selected based on pre-policy outcome levels, I also estimate whether pre-treatment outcomes predict treatment status and find either no significant relationship or one in the opposite direction. Third, the large influx of students and staff resulting from this policy can alter the composition of primary and secondary schools, potentially driving my results. I investigate whether the shares of white students, students who attended only public schools, and students whose mothers completed at least high school are changing as a result of this policy. I find no evidence that changes in composition could explain the observed results.

This paper contributes to the broader literature on the effects of college expansion or proximity on educational attainment, which primarily focuses on the direct impact through college enrollment and completion (Card, 1993; Currie and Moretti, 2003; Noghanibehambari and Tavassoli, 2022; Belskaya et al., 2020). My paper contributes to the evidence that future educational opportunities shape households' early educational decisions and, thus, play a significant role in increasing overall human capital accumulation, a result especially relevant for low- and middle-income countries where average attainment falls well below high school completion. I directly build on and contribute to the literature examining the downstream effects of subsequent educational opportunities on earlier educational attainment. While a strand of this literature examines the effects of affirmative action on secondary students' outcomes (Akhtari et al., 2023; Khanna, 2020), my paper more closely aligns with a growing literature on the impact of expanding the supply of higher education institutions on lower-level students' educational attainment (Bedard, 2001; Jagnani and Khanna, 2020; Katzkowicz et al., 2023; Rands and Barsanetti, 2025; Lallemant and Dillon, 2026).

I build on this literature in two ways. First, I provide novel evidence that students shift pre-tertiary educational decisions in response not only to elite and traditional four-year universities, but also to technical higher education, such as two-year trade schools. Second, through my framework and empirical analysis of mechanisms, I can identify the channels through which higher education expansion can affect high school completion, and the necessary conditions for these effects to take place. In particular, I formally highlight the role of high returns to tertiary schooling in increasing expected returns to high school, especially in settings where the latter are modest relative to the former. This framework explains competing evidence in the literature, where previous work by Bedard (2001) has found decreases in high school completion in the United States following college expansion, while evidence in LMICs points in the opposite direction (Jagnani and Khanna, 2020), and provides evidence relevant to policy-makers on where these downstream effects can be expected.

The remainder of this paper is organized as follows. Section 2 details the expansion policy. Section 3 presents a theoretical framework to rationalize how students' lower-level educational decisions are affected by the expansion of tertiary education. Section 4 describes the data and provides more detail on the placement of HEIs. Section 5 describes the method used in this paper. Section 6 reports the results for all pre-tertiary outcomes. Section 7 provides empirical evidence on the mechanisms through which the policy operates. Section 8 presents robustness analyses, and Section 9 concludes with a discussion of the results.

2 Background

2.1 Higher education expansion program in Brazil

Tertiary education in Brazil comprises public and private institutions. The former is primarily provided by the federal government through Federal Universities and Trade Schools.[3] These institutions are free to students, and admissions are primarily based on a national exam (ENEM) test score that students completing high school can take once a year.[4] After obtaining their college-entry exam test scores, students apply to federal universities through the Unified Selection System, SISU, a centralized platform that allows students to choose the major and HEI combination they wish to pursue. Private and state universities have their own admission process, although some also use ENEM test scores.

In 2003, the federal government of Brazil launched a large higher education expansion program, with the main goal of reducing the concentration of higher education institutions in higher-income regions and capitals, and using higher education to promote regional development, aiming to increase municipalities' income and reduce regional inequalities. The government selected municipalities based on geographical and economic criteria, such as the municipality's Human Development Index (HDI), population size, local economic activities, and pre-existing supply of higher education

[3]State governments can also create and fund state universities, which are also free to students.

[4]The national unified exam, ENEM, started being used in college admissions in 2010. Beforehand, students took individual exams for each university they would apply to. Trade schools have alternative admission routes, but also use ENEM as part of their admissions process.

institutions (Barbosa et al., 2020; Ministério da Educação [MEC], 2011). The policy included the expansion of new public and free universities or trade schools, as well as new campuses of already existing HEIs, and was divided into three phases.

The first expansion phase, called Expansion I, occurred between 2003 and 2007, and its main goal was to expand higher education institutions into municipalities in the inner part of the country, thus earning the name "interiorization phase". During this period, 98 institutions were created. The second expansion phase, called REUNI, occurred between 2008 and 2012 and combined the continuation of the expansion started in 2003 with the restructuring of existing universities, along with expanded enrollment capacity and infrastructure improvements, including new buildings and classrooms. In this second phase, 247 new institutions were created.

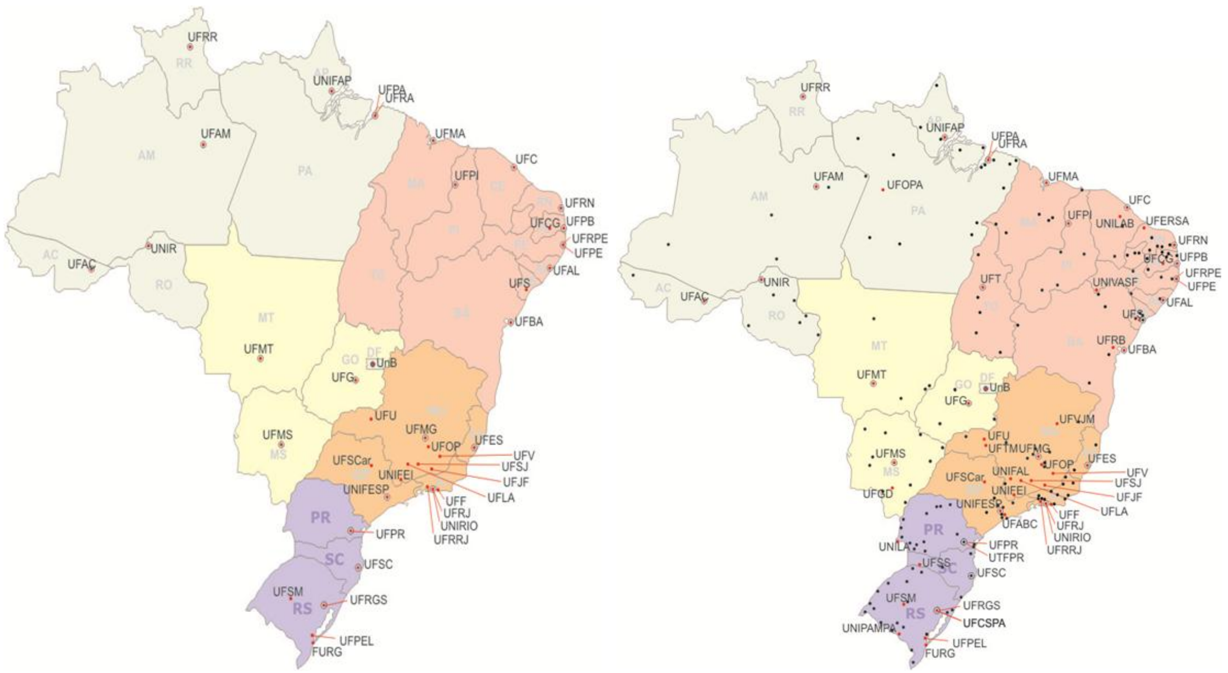
The third phase of the expansion occurred between 2012 and 2014, although institutions continued to open up until 2017. It focused on coupling higher education expansion with initiatives to develop underserved regions and on launching an array of public programs, such as policies to expand medical school enrollment and to include people with disabilities in higher education. All these additional policies targeted college students enrolled in these newly opened institutions. During this phase, 239 HEIs were created.

Throughout these three phases, 554 municipalities (nearly 10% of all Brazilian municipalities) received at least one institution between 2003 and 2017. Of those, 252 municipalities received their first higher education institution. Figure 1 presents the distribution of federal university campuses before and after the expansion program and illustrates the policy's goal of decreasing the concentration of higher education institutions in capitals, metropolitan areas, and more developed regions of the country.

2.2 Federal trade schools

Trade schools account for approximately two-thirds of the HEIs opened under the federal expansion program. These institutions differ from universities in relevant ways that could generate differential effects, depending on the type of HEI a municipality receives. Trade schools are part of the professional and technological education programs in Brazil and were first conceived in the early 1900s as schools to provide lower-income individuals with technical skills, such as carpentry. Between 1930 and 1970, the federal government expanded the supply of technical education to meet the growing demand for middle-skilled labor arising from industrialization, particularly for employment in the automobile and petrochemical industries. At this time, trade schools started

Figure 1: Expansion of Federal Universities from 2002 (left) to 2014 (right)



Note: This figure comes from the Brazilian Ministry of Education website. Distribution of federal universities in Brazil in 2002 (left), with 45 institutions in 141 municipalities, with a total of 151 campuses, and in 2014 (right), with 63 institutions in 275 municipalities, with a total of 321 campuses. Colors represent the different regions of Brazil. This map does not include Trade Schools. Before the policy, there were 140 federal trade school campuses in Brazil.

providing a high school-equivalent education to attract young individuals (Faveri et al., 2018).

In the 1980s and 1990s, trade schools began offering training in computational skills and information technology (IT) to keep pace with changes in labor demand. Up until 2000, they remained focused on high school-equivalent education and short-term technical courses targeting hard skills. They were then incorporated into the Brazilian Higher Education system, although institutes still offer technical training that does not require a high school diploma. Since then, trade schools offer three main types of courses: integrated technical courses, in which students are enrolled in high school and a technical course simultaneously, and which last 3 to 4 years; subsequent technical courses, with a minimum duration of 2 years, in which students need to have a high school diploma to enroll; and undergraduate majors, which are similar to traditional universities[5]. Trade schools can also offer short-term courses, such as elderly caregiving and basic English, as well as graduate education, such as a Master's in Materials Engineering.

[5] Examples of integrated technical courses are IT and construction. Examples of subsequent technical courses are business administration, workplace safety, and nursing technician. Examples of undergraduate majors are energy engineering, architecture, and math.

3 Why and when do higher education opportunities matter? A theoretical framework

Given that the provision of higher education does not directly affect the availability or costs related to secondary schooling, it is important to understand when and why an increased supply of subsequent educational opportunities can change earlier educational decisions. In this section, I develop a simple model to explain how students decide whether to complete high school and how the availability of higher education factors into this decision. The goal of this model is twofold: first, I will identify the channels through which the availability of higher education influences the decision to graduate from high school, and the specific parameters that may be affected by an increased supply of local HEIs. Second, I will identify the necessary conditions that need to be in place for trickle-down educational effects to occur, which will be key to reconciling mixed evidence stemming from different contexts, such as Jagnani and Khanna (2020) for India and Bedard (2001) for the United States, and generate predictions of what can be expected in the Brazilian setting.

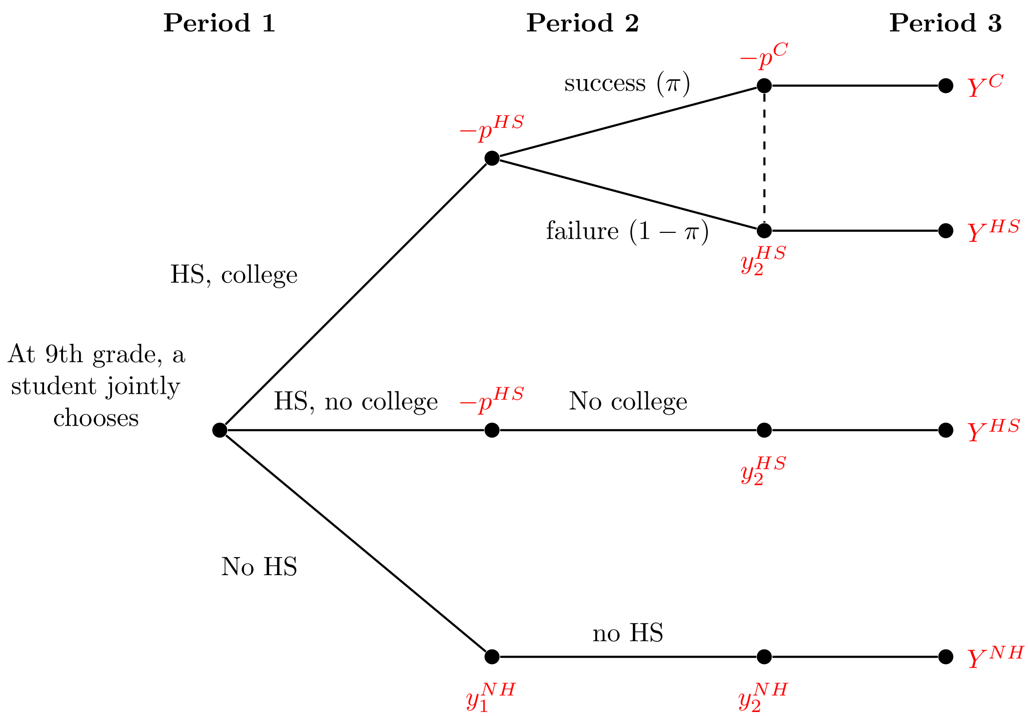
3.1 A student's decision: baseline model

Figure 2 illustrates the decision of a forward-looking student who has just finished primary education (9th grade in the Brazilian system). At that point, a student must simultaneously decide whether to continue to high school and graduate ($s = 1$) and, upon doing so, whether to *attempt to* continue to higher education ($d = 1$). Three possible paths are available to a student. If a student decides to drop out of high school ($s = 0$), they automatically choose not to pursue higher education ($d = 0$) and thus follow the (no high school, no college) path and immediately join the labor force in period 1. On the other hand, if a student chooses to graduate from high school ($s = 1$), they will also choose whether to pursue higher education. If choosing to do so, a student will then be on the (high school, college) path. Otherwise, they will follow the (high school, no college) path and join the labor force in period 2, immediately after completing high school. If a student chooses to follow the (high school, college) path, they will successfully enroll in and graduate from higher education with probability π_c, and fail and join the labor force with a high school degree with probability $(1-\pi_c)$, similarly to those who chose the (high school, no college) path[6]. This uncertainty is key to modeling the Brazilian setting, where public higher education spots are limited, acceptance rates

[6]In this framework, I will abstract away selection issues that would lead high school-level earnings to be different for a student who chose high school and no college, and a student who chose high school, college but failed to progress to higher education.

are below 10%, and higher education financing for private higher education is rare for most of the policy period. Thus, lower-income public school students, the focus of this paper, face significant constraints in attending higher education and might fail to do so even after choosing to try.

Figure 2: Decision tree with joint choice of high school and college attendance



In this setting, both high school and college attendance incur transportation/relocation and material costs, and a student must pay p^{HS} and p^C if attending high school in period 1 and higher education in period 2, respectively. For simplicity, I abstract from tuition-related costs. Since public high school is readily available and HEIs opening as part of the Brazilian policy are free, this is a reasonable abstraction[7]. At any point when a student enters the labor force, they will start earning constant wages specific to their final educational level: y^{NH} for dropouts, y^{HS} for high school graduates, and y^C for higher-education graduates. After the initial schooling choice and the realization of the college lottery, their educational attainment is fixed, and a student will earn the same wage for the remainder of their life, here represented by period 3.

In a setting where higher education is not available or where distance and moving costs are so high for the median student that college is not in their choice set, students will choose only

[7]Private higher education is also a relevant share of tertiary institutions in Brazil. For simplicity and due to data availability, I do not directly model private HEI enrollment as a student outcome. For my sample, which consists only of public school students coming from lower-income households, and for most of this time period, when loans to attend private colleges were limited, this is an acceptable simplification.

between (high school, no college) and (no high school, no college), or simply whether to graduate high school. In this case, if the high school premium is high enough to justify the foregone earnings in period 1, students will choose to stay in school rather than drop out and join the labor force. Now consider a world where higher education is available. In addition to the high school decision, a 9th-grader will simultaneously choose whether to pursue higher education after completing high school. In this scenario, choosing to graduate high school opens an additional path, (high school, college), which might lead to greater lifetime earnings.

The decision to attend high school varies greatly depending on which world a student is in. Assuming students have certainty about wages by education level, when higher education is not an option, the expected earnings from graduating high school are simply the high school wage, and the schooling decision is a simple comparison of lifetime earnings with and without graduating high school. On the other hand, with the added possibility of pursuing higher education (and thus accessing higher-education wages), students now face uncertainty about their future earnings when choosing the path corresponding to (high school, college). Or: $\mathbb{E}[y^{HS}_{hs,nocol}] = y^{HS}$, while $\mathbb{E}[y^{HS}_{hs,col}] = (1-\pi_c)y^{HS} + \pi_c y^C$. Simply put, this additional educational opportunity represents a potential increase in the expected returns to high school completion, provided that wage levels vary considerably by educational level in the local labor market.

I formalize the decision-making process illustrated in Figure 2 in a simple three-period problem that shows how higher education opportunities can shift a student's decision to attend high school through different channels. Consider the maximization problem below, where, at the end of 9th grade (period 1), a forward-looking student chooses a schooling path $(s,d) \in \{(0,0),(1,0),(1,1)\}$, as defined above. Because graduating from college requires first graduating from high school, the pair $(0,1)$ is not feasible. The student maximizes the expected utility:

$$\max_{(s,d)\in\{(0,0),(1,0),(1,1)\}} U(C_1) + \beta_2\,\mathbb{E}\big[U(C_2)\big] + \beta_3\,\mathbb{E}\big[U(C_3)\big], \tag{1}$$

where u is a standard strictly increasing and strictly concave utility function and $\beta_2 = \beta^3$ and $\beta_3 \equiv \sum_{t=8}^{N} \beta^{t-1}$. β is the student's discount factor. The consumption streams for each period are:

Period 1. Period 1 lasts 3 years, the duration of high school in Brazil, grades 10 to 12:

$$C_1 = \bar{k}_1 + (1-s)\,y_1^{NH} - s\,p^H,$$

where $\bar{k}$ is the level of household assets, which includes the earnings of other family members, $\bar{k}_1 = 3\bar{k}$ and $y_1^{NH} = 3y^{NH}$.

Period 2. Period 2 lasts 4 years, the typical duration of tertiary education in Brazil:

$$C_2 = \begin{cases} \bar{k}_2 + y_2^{NH} & \text{if } (s,d) = (0,0), \\ \bar{k}_2 + y_2^{HS} & \text{if } (s,d) = (1,0), \\ \bar{k}_2 + y_2^{HS} & \text{if } (s,d) = (1,1) \text{ and not admitted with probability } (1-\pi_c), \\ \bar{k}_2 - p^c & \text{if } (s,d) = (1,1) \text{ and admitted with probability } \pi_c. \end{cases}$$

where $\bar{k}_2 = 4\bar{k}$, $y_2^{NH} = 4y^{NH}$ and $y_2^{HS} = 4y^{HS}$.

Period 3. Period 3 summarizes the remainder of an individual's working life as the present value of the per-year consumption C_3. After period 2, an individual's education level is fixed, and consumption will be constant for the following N periods during which the individual remains employed.

$$C_3 = \begin{cases} \bar{k} + y^{NH} & \text{if } (s,d) = (0,0), \\ \bar{k} + y^{HS} & \text{if } (s,d) = (1,0), \\ \bar{k} + y^{HS} & \text{if } (s,d) = (1,1) \text{ and not admitted}, \\ \bar{k} + y^{C} & \text{if } (s,d) = (1,1) \text{ and admitted}, \end{cases}$$

For simplicity, I assume $\bar{k}$ to be constant over time. Similarly, earnings by level of education are constant and known to students at the time of the schooling decision, and students' final educational attainment cannot change after this initial decision and the realization of the college draw. For paths $(0,0)$ and $(1,0)$, there is no uncertainty given that the student will not apply to college, and the only source of uncertainty is the college admission probability, $\pi_c \in (0,1)$.

In period 1, a student who chose to attend high school, $s = 1$, either by following the $(1,0)$ or $(1,1)$ paths, will incur costs $p^{HS} > 0$, while a dropout will immediately join the labor market and earn total wages y_1^{NH}. In period 2, a student who chose $(1,0)$ will finish high school and join the labor force earning wages y_2^{HS}, while a student who chose $(1,1)$ will continue to higher education with probability $\pi_c \in (0,1)$ and incur costs $p^C > 0$. If failing to access college with probability $1-\pi_c$, the student will then join the labor force and earn wages y_2^{HS}, similarly to those individuals choosing $(1,0)$. In period 3, those who proceeded to higher education in period 2 will earn a wage of y^C for the N years in which they will remain active in the labor force, whereas those who stopped at high school and primary education will earn y^{HS} and y^{NH}, respectively. For simplicity, I will assume that costs are drawn independently and uniformly: $p^H \sim U[0, \bar{P}^H]$ and $p^c \sim U[0, \bar{P}^c]$. I further assume that $\bar{P}^H < \bar{k}_1$ and $\bar{P}^c < \bar{k}_2$ so that period-1 and period-2 consumption are strictly positive on all paths.

A student will decide whether to follow $(0,0)$, $(1,0)$, or $(1,1)$ by comparing the expected lifetime utility they will receive from each path. Assuming $U(c) = \ln c$, the lifetime utilities of the three paths are:

$$V(0,0) = \ln(\bar{k}_1 + y_1^{NH}) + \beta_2 \ln(\bar{k}_2 + y_2^{NH}) + \beta_3 \ln(\bar{k} + y^{NH}), \tag{2}$$

$$V(1,0) = \ln(\bar{k}_1 - p^H) + \beta_2 \ln(\bar{k}_2 + y_2^{HS}) + \beta_3 \ln(\bar{k} + y^{HS}), \tag{3}$$

$$\begin{aligned} V(1,1) = {} & \ln(\bar{k}_1 - p^H) + \beta_2 \big[(1-\pi_c)\ln(\bar{k}_2 + y_2^{HS}) + \pi_c \ln(\bar{k}_2 - p^c)\big] \\ & + \beta_3 \big[(1-\pi_c)\ln(\bar{k} + y^{HS}) + \pi_c \ln(\bar{k} + y^C)\big]. \end{aligned} \tag{4}$$

By setting $V(1,0) = V(0,0)$, $V(1,1) = V(0,0)$ and $V(1,1) = V(1,0)$, I derive thresholds for when a student will prefer $(1,0)$ over $(0,0)$, $(1,1)$ over $(0,0)$ and $(1,1)$ over $(1,0)$, respectively, as shown in the Appendix. Combining the three thresholds, the student's decision rule is:

$$(s,d)^* = \begin{cases} (1,1) & \text{if } \big[p^H \le \bar{p}^H_{10} \text{ and } p^c \le \bar{p}^c_{11|10}\big] \text{ or } \big[p^H > \bar{p}^H_{10} \text{ and } p^c \le \bar{p}^c_{11|00}\big], \\ (1,0) & \text{if } p^H \le \bar{p}^H_{10} \text{ and } p^c > \bar{p}^c_{11|10}, \\ (0,0) & \text{if } p^H > \bar{p}^H_{10} \text{ and } p^c > \bar{p}^c_{11|00}. \end{cases} \tag{5}$$

where:

$$\bar{p}^H_{10} = \bar{k}_1 - (\bar{k}_1 + y_1^{NH}) \left(\frac{\bar{k}_2 + y_2^{NH}}{\bar{k}_2 + y_2^{HS}}\right)^{\beta_2} \left(\frac{\bar{k} + y^{NH}}{\bar{k} + y^{HS}}\right)^{\beta_3}. \tag{6}$$

$$\bar{p}^c_{11|10} = \bar{k}_2 - (\bar{k}_2 + y_2^{HS}) \left(\frac{\bar{k} + y^{HS}}{\bar{k} + y^C}\right)^{\beta_3/\beta_2}. \tag{7}$$

$$\bar{p}^c_{11|00} = \bar{k}_2 - \left[\left(\frac{\bar{k}_1 + y_1^{NH}}{\bar{k}_1 - p^H} \right)^{1/\beta_2} \frac{\bar{k}_2 + y_2^{NH}}{(\bar{k}_2 + y_2^{HS})^{1-\pi_c}} \left(\frac{\bar{k} + y^{NH}}{(\bar{k} + y^{HS})^{1-\pi_c} (\bar{k} + y^C)^{\pi_c}} \right)^{\beta_3/\beta_2} \right]^{1/\pi_c} . \tag{8}$$

Each threshold marks the point where a student is indifferent between two options: at $p^H = \bar{p}^H_{10}$, a student is indifferent between $(0,0)$ and $(1,0)$; at $p^c = \bar{p}^c_{11|10}$, a student is indifferent between $(1,1)$ and $(1,0)$; and at $p^c = \bar{p}^c_{11|00}$, a student is indifferent between $(1,1)$ and $(0,0)$. In the absence of higher education, the relevant threshold would only be $\bar{p}^H_{10}$, which mediates the choice between (no high school, no college) and (high school, no college) and simply compares the value of going to high school versus dropping out. With the added higher education opportunity, it is now possible for students who would drop out of high school via $\bar{p}^H_{10}$ to choose the (high school, college) path through the $\bar{p}^c_{11|00}$ threshold: for those with prohibitive p^H to choose high school alone, high school now becomes attainable when $p^c \leq \bar{p}^c_{11|00}$.

Let S^* denote the aggregate share of students who graduate high school. A student graduates high school either by choosing $(1,0)$ (so $p^H \leq \bar{p}^H_{10}$) *or* $(1,1)$ over $(0,0)$ (so $p^H > \bar{p}^H_{10}$ and $p^c \leq \bar{p}^c_{11|00}$). Hence:

$$S^* = \Pr(p^H \leq \bar{p}^H_{10}) + \Pr\big(p^H > \bar{p}^H_{10} \text{ and } p^c \leq \bar{p}^c_{11|00}\big). \tag{9}$$

Applying the law of total probability and independence of p^H and p^c:

$$S^* = \Pr(p^H \leq \bar{p}^H_{10}) + \int_{\bar{p}^H_{10}}^{\bar{P}^H} \Pr\big(p^c \leq \bar{p}^c_{11|00}(p^H)\big) \cdot f(p^H)\, dp^H,$$

where $f(p^H) = 1/\bar{P}^H$. Under $p^c \sim U[0, \bar{P}^c]$, $\Pr(p^c \leq \bar{p}^c_{11|00}(p^H)) = \bar{p}^c_{11|00}(p^H)/\bar{P}^c$. Therefore:

$$S^* = \theta_H + \frac{1}{\bar{P}^H \bar{P}^c} \int_{\bar{p}^H_{10}}^{\bar{P}^H} \bar{p}^c_{11|00}(p^H)\, dp^H. \tag{10}$$

The first term represents the share that finds high school worthwhile on its own, while the second is the share that would not attend high school alone but is induced to complete the full college path when college costs are low enough. Note that S^* does *not* depend on $\bar{p}^c_{11|10}$, the threshold pertaining to students who move from $(1,0)$ to $(1,1)$.

3.2 Necessary conditions for pre-tertiary attainment to increase

Given the composition of S^* described above, changes in the share of students completing high school can stem from two distinct transitions. The first moves students from (no high school, no college) to (high school, no college). The threshold which mediates this transition, $\bar{p}^H_{10}$, depends only on household assets and income, $\bar{k}$, and wage levels, y^{NH} and y^{HS}. Importantly, it does not depend on parameters related to higher education opportunities that might be directly affected by HEI expansion, such as college attendance costs, p^C, and the likelihood of attending an HEI, π_c. Thus, movements from $(0, 0)$ to $(1, 0)$ will result solely from changes in the local labor market due to HEI openings rather than from changes in educational opportunities. While these movements do not speak to the role of future educational opportunities directly, it is worth examining whether and when we can expect the policy to alter the high school decision while not changing the availability of higher education opportunities, as Jagnani and Khanna (2020) and Lallemant and Dillon (2026) have documented important infrastructure and labor market changes in places receiving new HEIs in India and Uganda, respectively.

The second relevant transition moves students from the (no high school, no college) path directly into (high school, college), and operates through the $\bar{p}^c_{11|00}$ threshold. Equation 8 shows that $\bar{p}^c_{11|00}$ depends not only on earnings by education levels and household wealth, but also on college attendance costs (via the support $\bar{P}^c$) and the probability of enrolling in higher education, π_c. These movements, although also affected by changes in the local labor market, are more closely related to the policy's direct effect. Here, the college decision directly alters the high school decision by changing cost and access parameters related to higher-education attendance. If we assume that, at least in the short run, local wage levels do not change following the opening of an HEI, all movements in S^* would stem from students moving from $(0, 0)$ to $(1, 1)$ and operate through an increase in educational opportunities. While I will empirically assess whether this is true, for completeness, I will allow wages to change with the policy and examine the implications of these changes on schooling decisions.

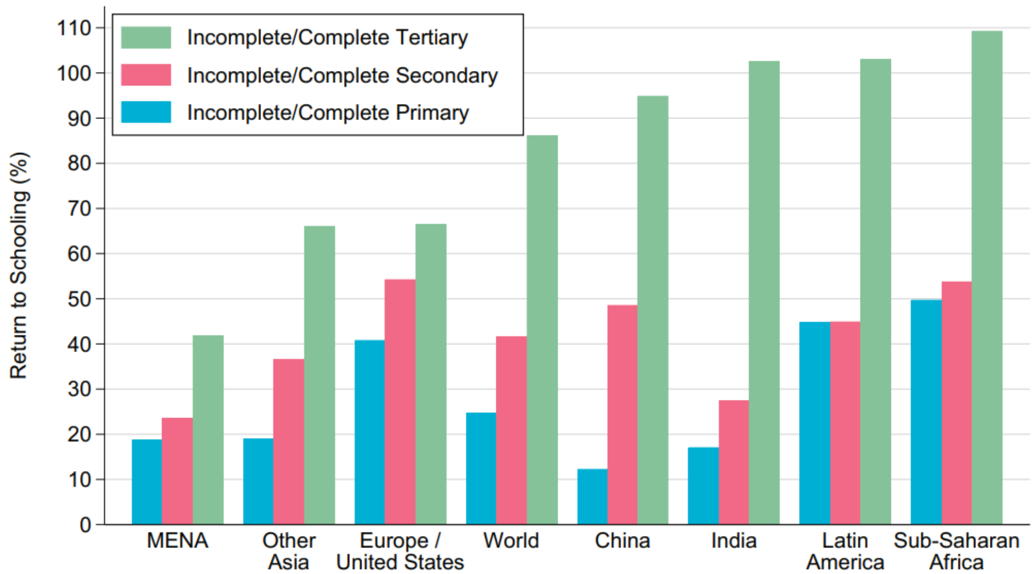
Three main conditions need to be in place for the higher education expansion to increase secondary school completion rates. First, a necessary condition for both transitions described above is that there is a mass of students who chose not to graduate from high school in the absence of higher education. That is, at least some students were choosing the path (no high school, no college) prior to the policy. Since students will choose this path only when $p^H > \bar{p}^H_{10}$, this translates into a condition on the threshold $\bar{p}_{10}^{H}$: $\bar{p}_{10}^{H} < \bar{P}^{H}$. As shown in the Appendix, this condition fails whenever high school wages tend to infinity. In practice, this implies that in places where the high school premium is very high, students will always choose to graduate from high school, and new HEIs will have no one to move into either of the two high school paths. This necessary condition then imposes an upper bound on the high school wage premium.

The second condition pertains only to movements from (no high school, no college) to (high school, no college). For students to be moved to (high school, no college), the threshold governing that transition, $\bar{p}_{10}^{H}$, needs to be positive, so that $p^{H} < \bar{p}_{10}^{H}$ for at least some students, given that $p^{H} > 0$. A necessary condition for $\bar{p}_{10}^{H} > 0$, as shown in the Appendix, is that high school wages must be larger than primary-level wages. This is needed so completing high school alone can be preferred over dropping out. These two conditions combined state that, for students to be moved into the (high school, no college) path by the policy, high school wages need to exceed primary education wages, but not so much that all students would have chosen (high school, no college) to begin with.

The third condition pertains only to transitions directly into the (high school, college) path. Similar to the condition above, for students to transition from (no high school, no college) to (high school, college), it must be true that $p^{c} < \bar{p}_{11|00}^{c}$ for at least some students. Since $p^{c} > 0$, this implies that $\bar{p}_{11|00}^{c}$ must be strictly positive. In the appendix, I show that this condition implies that higher education earnings need not only to be larger than high school earnings, but also above a certain threshold, $\bar{y}^{C}$. Together with the first stated condition, this implies that for students to choose to graduate high school because of college opportunities, the college wage premium must be sufficiently large, while high school earnings cannot greatly exceed primary education returns.

Figure 3 shows substantial variability in returns to schooling by education level across regions. In places like India, Latin America, and Sub-Saharan Africa, tertiary returns greatly exceed those for secondary schooling, whereas primary and high school returns are much closer in magnitude. On the other hand, in high-income countries such as Europe and the United States, high school returns are significantly higher than those for primary education, and tertiary education gains represent a much smaller jump than those observed in LMICs. According to this simple model and the necessary conditions described above, India, Latin America, and Sub-Saharan Africa would simultaneously have a much larger pool of high school dropouts to move, and a much larger college premium to leverage into pulling students from (no high school, no college) to the (high school, college) path.

Figure 3: Returns by region - Gethin (2025)



Note: This figure comes from Gethin (2025), published in the Quarterly Journal of Economics.

3.3 Comparative statics

In this section, I analyze how policy-driven changes in parameters might affect S^*, and through which of the two possible transitions described above. The opening of an HEI in a student's municipality might affect multiple aspects of a student's schooling decision. First, an expansion of local higher education supply might directly affect the costs of college attendance and the likelihood that a local student will be admitted. The former is a direct result of reduced transportation or relocation costs due to shorter distances to the nearest HEI. The latter, on the other hand, results from an increase in the number of available higher education spots in the municipality. While the change in costs is straightforward, in later sections I empirically assess whether the likelihood of admission and enrollment changes for local students following the opening of an HEI.

Second, in addition to generating new educational opportunities, newly established HEIs might change the local labor market in important ways that might affect schooling decisions. For instance, if new colleges create new job opportunities for staff or construction, demand for primary- and secondary-educated workers might increase, changing dropout and high school graduate wages. In addition, general equilibrium effects might arise given potential changes in educational attainment brought about by the policy. Whether and in which direction these changes take place is an empirical question I will assess in later sections. For now, I will show the direction I expect high school graduation rates to move given changes in wage parameters. Since S^* is monotonic in $\bar{p}^H_{10}$, $\bar{p}^c_{11|00}$, and $\bar{P}^c$, as shown in the Appendix, I can use these thresholds to draw comparative statics analysis.

3.3.1 Cost of attending higher education

Changes in the cost of attending higher education p^c only affect movements from $(0,0)$ to $(1,1)$ and, thus, $\bar{p}^c_{11|00}$. By decreasing the distance to the nearest HEI, the policy lowers the upper support of the distribution of p^c, $\bar{P}^c$, shifting students relative to the fixed threshold. Since shifts in attendance costs amount to shifts in the support $\bar{P}^c$, the effect of changing attendance costs on the share of students graduating high school can be described by the following:

$$\frac{\partial S^*}{\partial \bar{P}^c} = -\frac{1}{\bar{P}^H \bar{P}^{c2}} \int_{\bar{p}^H_{10}}^{\bar{P}^H} \bar{p}^c_{11|00}(p^H)\, dp^H < 0. \tag{11}$$

A reduction in college attendance costs shifts more students below $\bar{p}^c_{11|00}$, moving them from (no high school, no college) to (high school, college). This effect is larger when high school attendance costs are lower, as students driven to attend high school because of new higher-education opportunities will have to bear upfront costs in period 1.

3.3.2 Likelihood of attending higher education

Similar to the cost effect, the likelihood of attending higher education, π_c, runs entirely through movements from no high school to the (high school, college) path. Changes in high school completion given changes in π_c are given by:

$$\frac{\partial S^*}{\partial \pi_c} = \frac{1}{\bar{P}^H \bar{P}^c} \int_{\bar{p}^H_{10}}^{\bar{P}^H} \frac{\partial \bar{p}^c_{11|00}(p^H)}{\partial \pi_c}\, dp^H > 0 \quad \text{when } \bar{p}^c_{11|00} > 0. \tag{12}$$

In places where the college premium is sufficiently high (i.e., the necessary condition $\bar{p}^c_{11|00} > 0$ is met), a higher probability of attending higher education unambiguously increases the share of students who graduate from high school. This effect is larger in places and among individuals for whom attendance costs, both at high school and in higher education, are lower. If the college premium is low such that $\bar{p}^c_{11|00} \leq 0$, changes in the probability of acceptance will not change local teenagers' pre-college educational attainment.

3.3.3 College-graduate wages

Changes in college-graduate wages, y^C, also operate solely by moving students from (no high school, no college) to (high school, college). As shown in the Appendix, I can differentiate $V(1,1) - V(0,0) = 0$ (the exact point where $p^c = \bar{p}^c_{11|00}(p^H))$) to sign $\partial \bar{p}^c_{11|00}(p^H)/\partial y^C$.

$$\frac{\partial \big[V(1,1) - V(0,0) \big]}{\partial y^C} = \frac{\beta_3 \pi_c}{\bar{k} + y^C} > 0, \tag{13}$$

so $\partial \bar{p}^c_{11|00}(p^H)/\partial y^C > 0$ for all p^H.

$$\frac{\partial S^*}{\partial y^C} > 0, \tag{14}$$

When college wages increase, students are more likely to complete high school via the (high school, college) path.

3.3.4 High-school-graduate wages

Changes in high school level wages, y^{HS}, affect both transitions, so changes in S^* caused by changes in y^{HS} will be a mix of students choosing to graduate high school through both (high school, no college) and (high school, college) paths.

Effect via transitions from (no high school, no college) to (high school, no college). Writing $R \equiv ((\bar{k}_2 + y_2^{NH})/(\bar{k}_2 + y_2^{HS}))^{\beta_2}((\bar{k} + y^{NH})/(\bar{k} + y^{HS}))^{\beta_3}$ and differentiating $\bar{p}^H_{10} = \bar{k}_1 - (\bar{k}_1 + y_1^{NH})R$ using the log-derivative of R:

$$\frac{\partial \bar{p}^H_{10}}{\partial y^{HS}} = (\bar{k}_1 + y_1^{NH})\, R \left[\frac{4\beta_2}{\bar{k}_2 + y_2^{HS}} + \frac{\beta_3}{\bar{k} + y^{HS}} \right] > 0. \tag{15}$$

If new HEIs increase local high school wages, high school enrollment becomes more attractive, raising $\bar{p}^H_{10}$ and moving students from $(0,0)$ to $(1,0)$.

Effect via transitions from (no high school, no college) to (high school, college). From (45) and (44) in the Appendix:

$$\frac{\partial \big[V(1,1) - V(0,0) \big]}{\partial y^{HS}} = (1 - \pi_c) \left[\frac{4\beta_2}{\bar{k}_2 + y_2^{HS}} + \frac{\beta_3}{\bar{k} + y^{HS}} \right] > 0, \tag{16}$$

so $\partial \bar{p}^c_{11|00}(p^H)/\partial y^{HS} > 0$ for all p^H.

An increase in high school wages increases the expected returns to the (high school, college) path, driving more students to choose (high school, college) over no high school. Note that changes in high school wages play a more important role when the likelihood of accessing higher education is lower. In the extreme, when a student expects to access higher education with certainty ($\pi_c = 1$), changes in high school wages will not affect high school completion decisions through the education opportunities channel.

Net effect.

$$\frac{\partial S^*}{\partial y^{HS}} > 0. \tag{17}$$

Increases in high school wages unambiguously raise high school completion through both margins: movements from (no high school, no college) to either (high school, no college) or (high school, college).

3.3.5 Primary-educated wages

Similar to high school wages, changes in primary-level wages will affect high school completion through both margins.

Effect via transitions from (no high school, no college) to (high school, no college). Using the product rule on $\bar{p}^H_{10} = \bar{k}_1 - (\bar{k}_1 + y_1^{NH})R$:

$$\frac{\partial \bar{p}^H_{10}}{\partial y^{NH}} = -R\left[3 + (\bar{k}_1 + y_1^{NH})\left(\frac{4\beta_2}{\bar{k}_2 + y_2^{NH}} + \frac{\beta_3}{\bar{k} + y^{NH}}\right)\right] < 0. \tag{18}$$

Effect via transitions from (no high school, no college) to (high school, college). y^{NH} enters $V(0,0)$ through $\ln(\bar{k}_1 + y_1^{NH})$, $\ln(\bar{k}_2 + y_2^{NH})$, and $\ln(\bar{k} + y^{NH})$:

$$\frac{\partial\big[V(1,1) - V(0,0)\big]}{\partial y^{NH}} = -\frac{3}{\bar{k}_1 + y_1^{NH}} - \frac{4\beta_2}{\bar{k}_2 + y_2^{NH}} - \frac{\beta_3}{\bar{k} + y^{NH}} < 0, \tag{19}$$

so $\partial \bar{p}^c_{11|00}(p^H)/\partial y^{NH} < 0$ for all p^H.

Net effect.

$$\frac{\partial S^*}{\partial y^{NH}} < 0. \tag{20}$$

Higher dropout wages unambiguously reduce the share of students who graduate from high school by making the dropout option more attractive and the foregone earnings more costly, thereby steering students away from both paths leading to high school graduation.

4 Data and sample characteristics

To obtain data on college openings, I construct a novel dataset using institutional documents and information provided on college websites. This dataset contains the opening year, the year authorization to operate was granted, municipality name, region and state, and the type of higher education institution (university or trade school) that opened as a part of the expansion program, ranging from 2003 to 2014. I define the treatment year as the year an HEI opened, and, in cases where this information is missing (15%), I replace it with the year authorization to operate was granted. I complement these data with the Higher Education Census data for 2002 and 2018, provided by the Ministry of Education, to obtain information on the number of HEIs pre- and post-policy and to determine whether receiving municipalities had a public or private institution before 2003.

For all of my analyses, I construct a panel dataset of municipalities, covering 2007 to 2019 for 9th-grade test scores, high school progression and graduation rates, and labor market outcomes, and 2010 to 2018 for college-going outcomes (share of students taking a college-entry exam, exam scores, and share who applied, were accepted and enrolled in an HEI). The year restrictions are due to data availability. For all of my pre-tertiary education and college-going outcomes, I restrict my sample to public school students at baseline, but allow them to move to private high schools as a possible response to the policy. This is a key sample restriction, since public school students, on average, are lower-income and less likely to migrate to pursue tertiary education outside their municipality, and would likely be the most affected by this policy.

For 9th-grade standardized test scores, I use data from the SAEB/Prova Brasil, a national test administered every two years. Test scores are available biennially from 1995 to 2019, but municipality identifiers are available only from 2007 onward. I thus restrict my test score analysis to municipalities treated between 2008 and 2014. For graduation and high school transition outcomes, I use the universe of K-12 student enrollment in Brazil from the 2007 to 2017 School Census. I build high school graduation outcomes from both 9th- and 10th-grade baselines: conditional on having enrolled in 9th/10th grade, does a student graduate on time or with up to a 2-year delay? High school transition uses the 9th grade baseline (last grade of primary school): conditional on enrolling in 9th grade, does the student enroll in 10th grade within 3 years? I then generate graduation and transition rates at the municipality level. My main outcome, high school completion rates for the 10th grade baseline, includes municipalities treated between 2008 and 2013, and high school transition rates includes municipalities treated between 2008 and 2014.

My preferred graduation outcome uses the 10th-grade baseline for a few reasons. First, to construct graduation outcomes from the enrollment data, I need to observe students long enough into the future to determine whether they have completed 12th grade. Restricting to a 10th-grade baseline allows me to keep as much of the sample as possible while still restricting my baseline to a period prior to a critical dropout point (between grades 10 and 11) in the Brazilian system. Second, conditioning on students who have already started their first year of high school implies that, for those students, supply-side constraints are unlikely to be binding. Therefore, it follows that dropping out of or graduating from high school is mostly mediated by demand-side factors, which are the focus of this paper.

For both 9th- and 10th-grade baseline graduation outcomes, a limitation of using enrollment data is that I must infer that students who reach 12th grade and then disappear from the K-12 enrollment system have graduated. To bound my estimates, I use 12th-grade yearly municipality dropout rates and present them in the Appendix. The results do not change, and, in some cases, the point estimates are larger. To avoid relying on imputed data, I use the outcome constructed solely from enrollment data as my main outcome.

College-going outcomes stem from the unified selection system, SISU, which has been widely used for admissions to public HEIs since 2010. This contains individual-level information on the applicant's college-entry exam scores, municipality of residence, school municipality, whether they were accepted into college using SISU, and whether they enrolled. I aggregate this information to generate municipality-level mean test scores, share of 17- to 25-year-olds taking the national college entry exam, and application, acceptance, and enrollment rates from 2010 to 2017. It is worth noting that not all HEIs use the national exam and SISU for admission. Thus, the acceptance and enrollment rates calculated from this dataset reflect college-going outcomes only for a specific subset of HEIs. However, nearly all schools opening as part of the expansion policy use SISU as their only form of admission, and these outcomes are therefore informative of whether students attend HEIs created by this policy. Due to data availability, I restrict my analysis of college-going to municipalities that received an HEI between 2011 and 2014. To analyze possible changes in the local labor market, I employ administrative data from the Relação Anual de Informações Sociais (RAIS - Annual Report of Social Information), which contains the universe of formal sector employees and provides information on wages and hours worked, along with demographic characteristics such as gender, race, and educational attainment, and spans 2007 to 2019.

I use data on municipality GDP per capita, the share of the agricultural sector in GDP, per capita education expenditure, Human Development Index (HDI), whether the local government is aligned with the federal government, the existence of an HEI pre-policy, and population size to generate pre-policy municipality-level controls. These data come from the IPEA data database and are discretized into quintiles to ensure common support in my estimation. In all of my estimations, I restrict my sample to treated municipalities and to controls located at least 45 kilometers from a treated municipality.

4.1 Placement of HEIs

As previously discussed, the placement of HEIs in this policy was not random. The federal government followed several criteria to select receiving municipalities, so treated units may differ from control units in many ways. Table 1 presents descriptive statistics on municipality characteristics, students' demographic characteristics, and pre-tertiary outcomes at baseline for treated and control municipalities, as well as for those receiving a trade school or a university.

Overall, treated municipalities have higher GDP per capita, are less agriculture-intensive, have a higher HDI, and lower K-12 education spending than control municipalities. Contrary to the federal government's stated criteria, 42% of treated municipalities also had a pre-existing HEI, compared with only 5% in the control group. Treated municipalities have a lower share of students who reported working outside of the household and a higher share of students whose mothers had completed High School, although a similar share of students had mothers who completed higher education. Public school 9th-graders in treated municipalities have higher Portuguese and Math test scores than those in the control group. However, both groups are below the national average on the standardized tests, a result of the public school student sample restriction. On the other hand, conditional on starting 10th grade, students in control municipalities were 4 percentage points more likely to graduate from high school.

Since municipality characteristics may vary by type of treatment received, columns 4 and 5 of Table 1 compare pretreatment characteristics of municipalities that received a university versus those that received a trade school. The latter are less agriculture-intensive and have slightly higher

Table 1: Pretreatment Characteristics by Treatment Status and Type of Treatment Received

	Treated (1)	Control (2)	P-value	Trade Schools (4)	Universities (5)	P-value
Municipality Characteristics						
GDP per capita (R)	6,428.41	4,682.67	0.00	6,403.90	6,161.99	0.18
Share of Agriculture in GDP	0.13	0.29	0.00	0.12	0.16	0.00
HDI	0.57	0.50	0.00	0.57	0.55	0.00
Mayor aligned with federal government	0.09	0.08	0.09	0.09	0.09	0.86
Education spending per capita (R)	138.10	238.16	0.00	137.54	155.33	0.00
Had a HEI before the policy	0.42	0.05	0.00	0.43	0.30	0.00
Student's Characteristics - 9th Grade						
White	0.35	0.34	0.70	0.35	0.34	0.67
Working	0.22	0.27	0.00	0.21	0.23	0.07
Female	0.55	0.55	0.71	0.55	0.56	0.04
Mother completed High School	0.31	0.26	0.00	0.31	0.31	0.73
Mother completed higher education	0.09	0.09	0.20	0.09	0.09	0.94
Outcomes at baseline						
Portuguese Scores - 9th Grade	-0.39	-0.47	0.00	-0.38	-0.40	0.55
Math Scores - 9th Grade	-0.17	-0.21	0.01	-0.16	-0.19	0.52
High School Graduation	0.55	0.59	0.00	0.55	0.53	0.09
Observations	379	2,180		305	46	

Note: Test scores are z-scores standardized with respect to the population mean for each year, which includes private schools in Brazil and, thus, is the reason why public schools' test scores for both treated and control groups are negative. High School Graduation is the share of students entering 10th grade in 2007 who completed high school within five years. 28 municipalities received both a university and a trade school, and are threfore left out of the trade school vs university breakdown.

HDI scores than those receiving a university. I do not find large differences in students' characteristics or pre-treatment outcomes between municipalities that received a trade school and those that received a university.

Table 2: Correlation Between Pre-Treatment Outcomes and Treatment Status

Independent Variable	Graduate HS on Time (1)	Eventually Graduate HS (2)	Start High School (3)	9th Grade Portuguese (4)	9th Grade Math (5)	Share Taking Exam (6)	College Exam Scores (7)
Dependent Variable							
Treated	-1.004***	-0.770**	0.552	0.237	0.059	0.980	-0.001
	(0.370)	(0.374)	(0.464)	(0.191)	(0.168)	(0.974)	(0.002)
Observations	2,402	2,402	2,400	2,543	2,543	1,423	1,344
Pre-treatment year	2007	2007	2007	2007	2007	2010	2010

Note: Each column reports the coefficient on the pre-treatment outcome from a separate probit of treatment status on that outcome and the full set of municipality controls. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

One possible concern is that treated localities were chosen based on pre-treatment outcomes. For instance, municipalities with high secondary graduation rates or test scores pre-policy could have received new HEIs at higher rates due to increased demand for tertiary education. To assess if this is the case, I analyze whether pre-treatment graduation and transition rates, test scores, and college exam outcomes predict treatment assignment by running a probit regression of treatment status on pre-policy outcomes and municipality-level controls used in the main analysis. Estimates are presented in Table 2. Lower high school graduation rates are correlated with a higher likelihood of being treated. Selection in this direction, if anything, would bias my estimates downward and thus should not be a concern. This relationship is also expected, given the policy's stated goal of serving underserved and underdeveloped regions. To assess differences in trends, I will present event-study estimates in the main results section.

I will deal with the observed differences between treated and control municipalities in two ways. First, in a difference-in-differences framework, control and treated units need to be comparable in trends but not in levels. I will show in the results section that there is little evidence of pre-trends, which supports the comparisons between these two groups. Second, I will control for the aforementioned federal government criteria and for per-capita educational expenditure to account for potential differences in pre-treatment school quality.

5 Methods

To estimate the effects of higher education expansion on local students' pre-tertiary education and college-going outcomes, I will compare treated versus untreated municipalities before and after an HEI's opening in a difference-in-differences setting. Equation 21 presents a two-way-fixed-effect estimation for that comparison.

$$y_{m,t} = \alpha + \beta ReceivedHEI_m \cdot Post_{t,m} + \gamma_t + \mu_m + \Pi \mathbf{X}_m + \epsilon_{m,t} \tag{21}$$

Where $y_{m,t}$ is the outcome of interest in municipality m at time t; $ReceivedHEI_m$ is the treatment variable, which equals one if the municipality received a higher education institution; $Post_{t,m}$ equals one if the period t is post-treatment in municipality m and zero otherwise; γ_t are year fixed effects to control for national trends in outcomes; μ_m are municipality fixed effects; and $\mathbf{X}$ is a vector of discretized pre-treatment municipality characteristics, which controls for factors taken into account by the Federal Government to assign treatment, such as share of agricultural sector, GDP per capita, presence of an HEI, population size, HDI, educational expenditures and alignment with the federal government[8]. The main coefficient of interest is β, which in this specification does not vary with time.

Since the policy is staggered and the treatment years in my analysis range from 2008 to 2014, I employ the staggered-timing difference-in-differences estimator proposed by Callaway and Sant'Anna (2021) and include municipality-level controls. This will allow for heterogeneous effects over time that should be relevant to this policy, as well as mitigate concerns about employing a Two-Way Fixed Effects estimator in a staggered treatment setting (Goodman-Bacon, 2021). In this approach, average treatment effects on the treated are calculated as described in Equation 22.

$$ATT(g,t) = E\left[\left(\frac{G_g}{\mathbb{E}[G_g]} - \frac{\frac{p_g(X)(1-G_g)}{1-p_g(X)}}{\mathbb{E}\left[\frac{p_g(X)(1-G_g)}{1-p_g(X)}\right]}\right) \cdot (Y_t - Y_{g-1})\right] \tag{22}$$

Where $ATT(g,t)$ is the average treatment effect measured at time t, $t \geq g$, for all municipalities that received an HEI starting in time g. $G_g = 1$ if a municipality is part of the group g, Y_t is the outcome for all units in the post-period of interest, t, and Y_{g-1} is the outcome for all units in the period immediately before when treatment started, $g-1$. $p_g(X) = P(G = g \mid X, G \in \{g, \text{control}\})$ is a propensity score calculated using a set of municipality-level covariates measured pre-policy that informs on the likelihood of belonging to cohort g conditional on covariates X and being either in cohort g or the control group. $\mathbb{E}[G_g]$ and $\mathbb{E}\left[\frac{p_g(X)(1-G_g)}{1-p_g(X)}\right]$ are normalizations to ensure that weights sum to one for both treated and control groups. ATTs for each time t and treated cohort g are then grouped following Equation 23, using the weighting function $w(g,t)$. This will allow us to estimate an overall average effect for the policy, θ.

$$\theta = \sum_{g \in \mathcal{G}} \sum_{t=2}^{\mathcal{T}} w(g,t) \cdot ATT(g,t) \tag{23}$$

My main identification assumption is that, in the absence of treatment, treated and untreated municipalities would have followed the same trends in the outcomes over time. Although this hypothesis cannot be formally tested, I will use event-study specifications as described in Equation 24 to provide evidence of no differential pre-trends. This specification will allow for the aggregation of ATTs across all treated cohorts, g, for each pre and post-policy period, ℓ. When $\ell < g$, the estimated ATT works as a placebo test that can provide suggestive evidence of whether the parallel

[8]Pre-tertiary outcomes are the likelihood of transitioning into high school, high school graduation, 5th grade Portuguese and Math scores, 9th grade Portuguese and Math scores, and average college-entry test scores. College-going outcomes are the share of 17- to 25-year-olds who take the college-entry exam, apply to college, are accepted, and enroll. Labor market outcomes are log wages by educational level.

trends assumption is met. Control municipalities are at least 45 kilometers away from a treated municipality and are never treated. Municipalities treated before the period for which data are available are excluded from my analysis.

$$\theta(\ell) = \sum_{g \in G} w(g, \ell)\, ATT(g, g + \ell), \quad \text{for each event time } \ell. \tag{24}$$

To address concerns that the non-random assignment of treatment may be correlated with outcomes, I will control for municipalities' characteristics that appear to have influenced the federal government's decision in selecting municipalities, as documented in institutional documents and the literature, in all specifications. With the inclusion of controls, the parallel trends assumption needs to hold conditional on municipality characteristics.

6 The impacts of higher education opportunities on earlier educational decisions

Table 3 shows the effects of higher education expansion on pre-tertiary outcomes, using Callaway and Sant'Anna (2021) and measured within 3 years of an HEI's opening. Panel A shows that receiving an HEI in a student's municipality does not affect the likelihood that 9th-graders transition to high school in the following three years. Conditional on being enrolled in 9th and 10th grade (columns 2 and 4), students are not more or less likely to graduate from high school on time. However, they are 1 percentage point more likely to graduate with up to a 2-year delay, which corresponds to a 2% increase relative to the pre-policy mean.

Panel B presents performance outcomes for 9th-graders and for students exiting high school who choose to take the college-entry exam. I find no changes in either 9th-graders' Portuguese and Math performance or college-entry exam scores. I do, however, find meaningful increases in the share of local young adults taking an optional college exam. The share of students writing the exam following the opening of an HEI increases by 2 percentage points, a 20% increase from the pre-policy mean. This increase in take-up does not seem to come at the expense of performance, as measured by college-entry exam scores.

Figure 4 shows that the effects on high school graduation increase over time, indicating that the downstream effects of higher education on pre-tertiary students' outcomes increase with the number of years a student is exposed to an HEI during their school career.

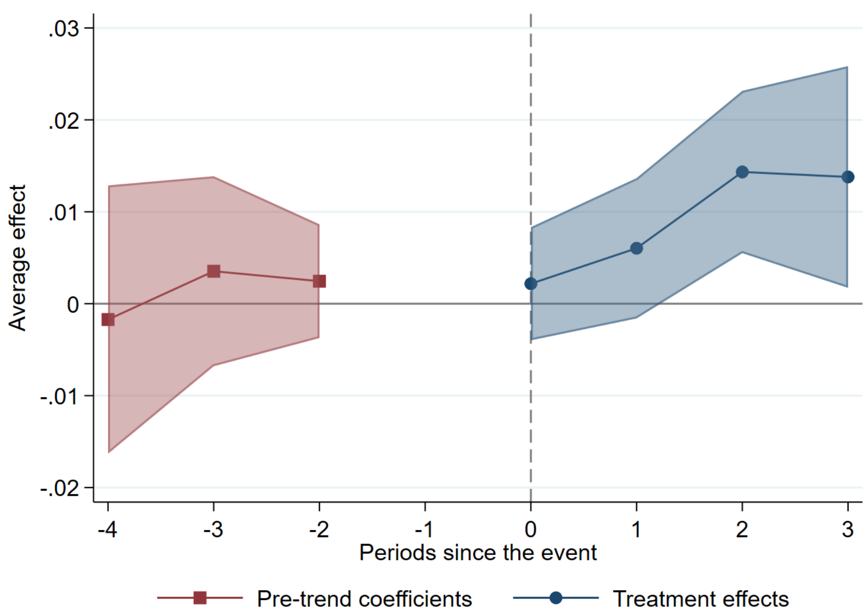
Table 3: Pre-Tertiary Outcomes

Panel A: *Persistence*		9th grade baseline		10th grade baseline	
	Start HS	Graduate on time	Eventually graduate	Graduate on time	Eventually graduate
	(1)	(2)	(3)	(4)	(5)
Treated	0.005	0.001	0.011*	0.000	0.010**
	(0.003)	(0.004)	(0.004)	(0.004)	(0.004)
Observations	19,208	19,208	14,410	21,616	16,814
Pre-policy Mean	0.79	0.40	0.49	0.46	0.54

Panel B: *Performance Outcomes*	9th Grade Math	9th Grade Portuguese	Share Taking Exam	College Exam Scores
	(1)	(2)	(3)	(4)
Treated	0.01	0.01	0.02***	-0.54
	(0.01)	(0.01)	(0.00)	(1.30)
Observations	17,548	17,548	16,485	16,485
Pre-policy Mean	-0.17	-0.39	0.10	525.48

Note: Standard errors in parentheses, clustered at the municipality level. All specifications include municipality-level control variables, measured pre-policy and discretized. ATTs are calculated using inverse probability weighting (IPW) and agregated to generate an overall average effect following Callaway and Sant'Anna (2021). Observations vary between outcomes due to years of data availability. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Figure 4: Likelihood of graduating High School with up to a 3-year delay



6.1 Does the type of Higher Education matter?

The positive effects above are comparable in magnitude with those in the Indian setting, as measured by Jagnani and Khanna (2020). Unlike that setting, however, the Brazilian policy not only expanded elite 4-year universities but also established new 4-year higher education institutions and 2-year trade schools. This policy's unique feature allows for testing whether other types of higher education opportunities can shift pre-tertiary students' schooling decisions. In this section, I examine whether the average effects measured above are driven by a specific type of HEI.

Although not statistically different, 9th graders in municipalities that received a university are 2 percentage points more likely to start high school in subsequent years, with no effect for students in localities that received a trade school. While not moving students into high school, trade schools do increase the likelihood of completing secondary schooling conditional on starting it. This suggests that the opportunity to access a trade school is more salient to students later in their schooling and who would have started high school in the absence of the policy. University openings, on the other hand, might change the composition of who starts high school in the first place. This difference in the composition of high school is further evidenced when comparing the 9th- and 10th-grade baseline estimates of the likelihood of eventually graduating (columns 3 and 5). When conditioning on 9th-grade enrollment, I find a large (but not statistically significant) 4% increase in the likelihood of eventually completing high school for students exposed to a university (column 3). However, this effect completely dissipates if conditioning on enrollment in 10th grade, likely due to changes in high school composition driven by the newly opened university.

6.2 Who benefits from higher education expansion?

Since all students in my sample are public school students, the overall effects are informative of how the policy affects lower-income, highly constrained students. However, within public schools, some students are more disadvantaged than others. In this section, I analyze whether the policy had heterogeneous effects across racial groups. Pre-policy, non-white students are 6 percentage points less likely to ever enroll in high school (70% versus 76% for white students) and 6 to 7 percentage points less likely to ever graduate (44% compared to 51% for the 9th-grade baseline, and 53% compared to 59% for the 10th-grade baseline). They are also lower-performing students, both on 9th-grade Portuguese and Math standardized tests and on college exam scores.

Table 5 shows that the policy disproportionately benefits non-white students. They are 2 percentage points more likely to ever enroll in high school (a 3% increase from the pre-policy mean), closing 1/3 of the pre-policy gap between white and non-white students. Although only marginally significant, estimates on the likelihood of ever graduating high school for non-white students are four times as large as those for white students: conditional on being enrolled in 10th grade, non-white students are nearly 1.5 percentage points more likely to eventually graduate high

Table 4: Heterogeneity by Type of HEI Received

Panel A: *Persistence*		9th grade baseline		10th grade baseline	
	Start HS	Graduate on time	Eventually graduate	Graduate on time	Eventually graduate
	(1)	(2)	(3)	(4)	(5)
Received a Trade School	0.003	-0.002	0.010*	-0.004	0.010*
	(0.003)	(0.005)	(0.005)	(0.004)	(0.004)
Observations	18,616	18,616	13,966	20,950	16,296
Pre-policy Mean	0.79	0.45	0.54	0.47	0.55
Received a University	0.016	0.011	0.021	-0.007	-0.003
	(0.009)	(0.010)	(0.012)	(0.008)	(0.010)
Observations	16,544	16,544	12,412	18,619	14,483
Pre-policy Mean	0.78	0.44	0.54	0.44	0.53
P-value (diff)	0.18	0.70	0.26	0.74	0.21
P-value (bootstrap)	0.17	0.28	0.41	0.64	0.24
Panel B: *Performance Outcomes*					
	9th Grade Math	9th Grade Portuguese	Share Taking Exam	College Exam Scores	
	(1)	(2)	(3)	(4)	
Received a Trade School	0.01	0.01	0.02***	-0.56	
	(0.01)	(0.01)	(0.00)	(1.44)	
Observations	17,032	17,032	18,702	17,425	
Pre-policy Mean	-0.16	-0.38	0.10	526.51	
Received a University	0.00	0.00	0.01	-2.32	
	(0.02)	(0.02)	(0.01)	(2.53)	
Observations	15,220	15,220	17,582	16,321	
Pre-policy Mean	-0.19	-0.40	0.11	516.85	
P-value (diff)	0.80	0.85	0.25	0.55	
P-value (bootstrap)	0.82	0.75	0.24	0.47	

Note: Standard errors in parentheses, clustered at the municipality level. All specifications include municipality-level control variables, measured pre-policy and discretized. ATTs are calculated using inverse probability weighting (IPW) and agregated to generate an overall average effect following Callaway and Sant'Anna (2021). Observations vary between outcomes due to years of data availability. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

school, a 4% increase from the pre-policy mean and 25% of the pre-policy gap relative to white students. Panel B shows that the increase in the share of students taking the college entry exam is driven solely by non-white students. Similar to the overall estimate, the effects on enrolling in and completing high school are increasing over time for non-white students, as shown in Figures 6 and 7, while effects for white students are small and not significant.

7 Channels through which higher education expansion affects pre-tertiary educational attainment

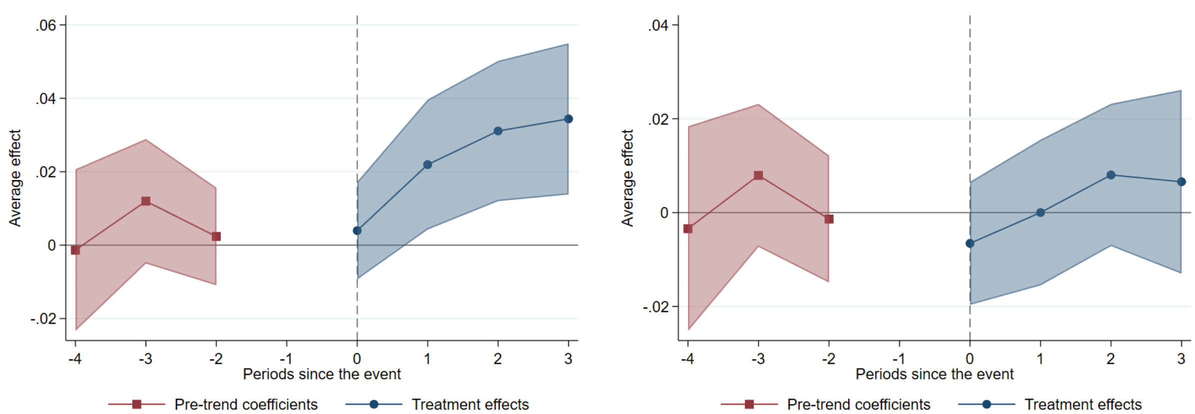
As highlighted in the model, changes in high school completion following the opening of an HEI can stem from two channels: the direct education opportunities channel, through changes in college

Table 5: Heterogeneity by Race

Panel A: *Persistence*		9th grade baseline		10th grade baseline	
	Start HS	Graduate on time	Eventually graduate	Graduate on time	Eventually graduate
	(1)	(2)	(3)	(4)	(5)
White	0.001	0.001	0.009	-0.006	0.005
	(0.007)	(0.008)	(0.010)	(0.007)	(0.008)
Observations	17,529	17,529	12,925	20,413	15,807
Pre-policy Mean	0.76	0.42	0.51	0.50	0.59
Non-white	0.021**	0.005	0.024**	0.004	0.020**
	(0.007)	(0.007)	(0.009)	(0.006)	(0.007)
Observations	17,529	17,529	12,925	20,413	15,807
Pre-policy Mean	0.70	0.35	0.44	0.42	0.53
P-value (diff)	0.05	0.70	0.26	0.27	0.18
P-value (bootstrap)	0.01	0.60	0.22	0.17	0.10
Panel B: *Performance Outcomes*					
	9th Grade Math	9th Grade Portuguese	Share Taking Exam	College Exam Scores	
	(1)	(2)	(3)	(4)	
White	0.01	0.01	0.00**	-0.01	
	(0.01)	(0.01)	(0.00)	(1.93)	
Observations	15,062	15,062	18,482	13,427	
Pre-policy Mean	-0.12	-0.35	0.05	533.43	
Non-white	0.00	0.00	0.01***	-1.56	
	(0.01)	(0.01)	(0.00)	(1.90)	
Observations	15,062	15,062	18,482	13,427	
Pre-policy Mean	-0.21	-0.42	0.05	526.15	
P-value (diff)	0.51	0.73	0.00	0.57	
P-value (bootstrap)	0.22	0.61	0.00	0.54	

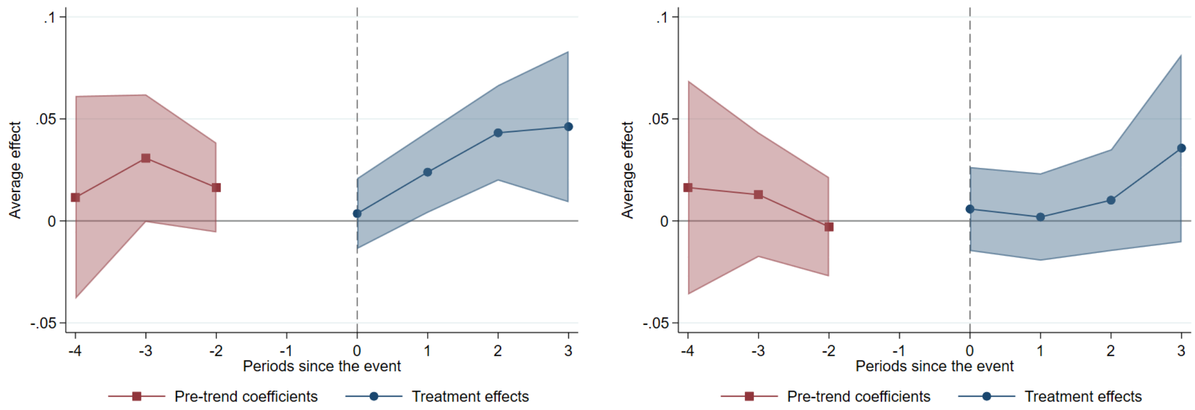
Note: Standard errors in parentheses, clustered at the municipality level. ATTs calculated following Callaway and Sant'Anna (2021). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Figure 5: Share of non-white (left) and white (right) 9th Graders who Start High School



attendance costs, p^C, and the likelihood of higher education admission, π_c; and the indirect labor market channel, through changes in local wages arising from general equilibrium effects and new

Figure 6: Likelihood of non-white (left) and white (right) 9th-graders to eventually graduate high school



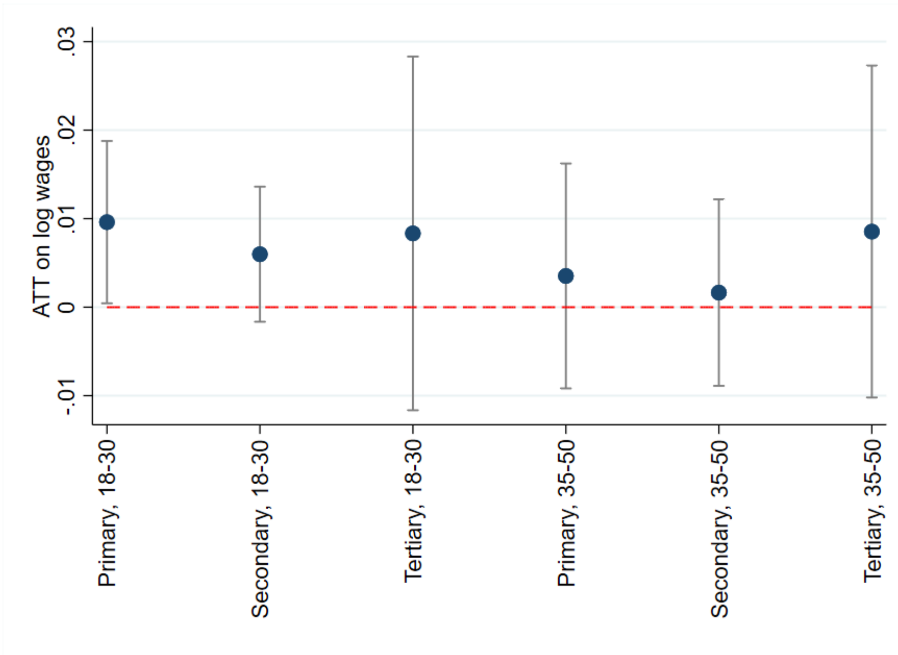
employment opportunities. In this section, I will first provide evidence on the mechanisms driving the observed effects by empirically examining which parameters change as a result of this policy. I will then empirically assess whether the necessary conditions derived in Section 3.2 hold.

7.1 The indirect labor market channel: changes in local wages

As documented by Jagnani and Khanna (2020) and Lallemant and Dillon (2026), the arrival of new HEIs can affect local infrastructure and the local labor market. If such changes move local wages in any way, they can also affect high school completion as predicted by my framework. If new institutions increase demand for low-skilled labor (e.g., construction jobs), wages for workers with only primary education, y^{NH}, might increase. Instead, if increases in labor demand focus on workers with high school or higher education degrees, then y^{HS} and y^{C} might increase. As derived in the comparative statics session, an increase in high school or college earnings, and a decrease in primary education wages could all be consistent with an increase in high school graduation. In this section, I assess whether the policy affected wages in the formal labor market in the short run (within 3 years of an HEI's opening) across different education levels and ages.

Changes in the local labor market do not seem to be driving the observed increases in high school completion. Point estimates are largest and significant only for primary-educated earnings among young workers, although all estimates shown in Figure 7 are indistinguishable from one another. If primary wages are, in fact, increasing as a result of the policy, the framework predicts that dropouts should increase, and high school graduation rates would move in the opposite direction of the observed effects. Therefore, if the labor market channel is at play, it operates by dampening

Figure 7: Effect on formal-sector wages by level of education and age



the policy's positive effects on pre-tertiary educational attainment rather than driving them.

Since labor market changes are the only channels through which movements from (no high school, no college) to (high school, no college) occur, it must be true that the observed positive effects stem solely from transitions directly into the (high school, college) path. The marginal students driving the results are those who would otherwise never graduate from high school but choose to do so because of increased higher-education opportunities.

7.2 The direct education opportunities channel

7.2.1 Probability of attending higher education

Although an increase in the supply of higher education is likely to raise a local student's probability of attending the new local HEI, this effect is not guaranteed. The national standardized test and unified higher education application system allow higher-income students from other municipalities to migrate and take advantage of newly created spots. In addition, spots at federal tuition-free HEIs are limited, and acceptance rates often range from 5 to 15%, since the application is nearly costless. To assess whether the policy increases higher education attendance for local public school students, I estimate the effect of receiving an HEI on the likelihood that local students apply to, are accepted to, and enroll in federal HEIs. Table 6 shows that local students are 2.5

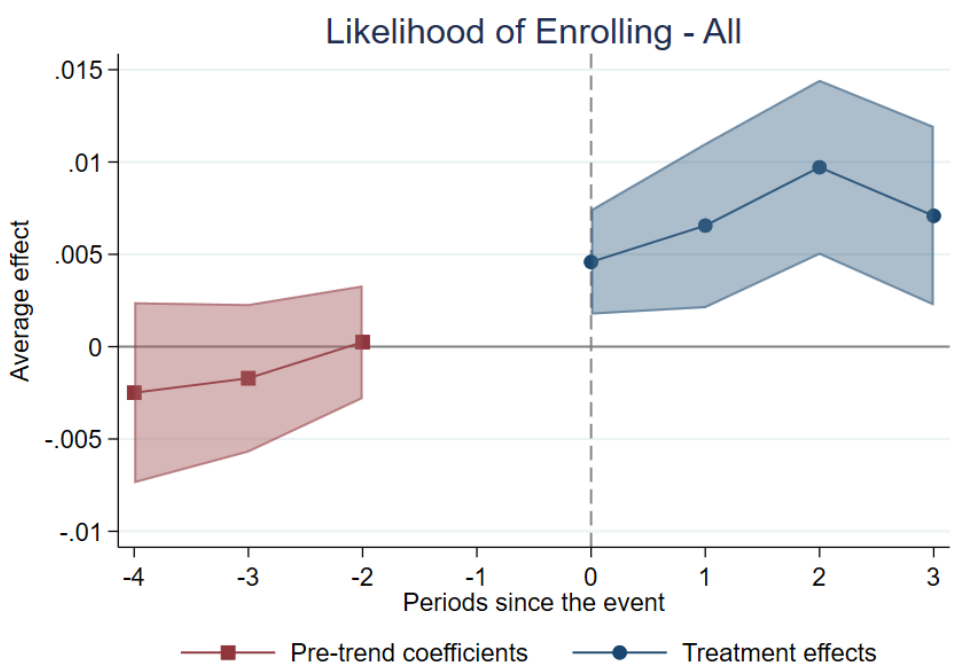
percentage points more likely to apply (14% increase relative to the pre-policy mean), 1.8 percentage points more likely to be accepted (nearly a 200% increase relative to the pre-policy mean), and 0.7 percentage points more likely to enroll (a 70% increase) in a public federal HEI following the policy. Enrollment effects are sustained over time, as shown by Figure 8, confirming that local students experience an increased probability of attending higher education (as represented by π_c in my framework).

Table 6: College-Going Outcomes

	Applied	Accepted	Enrolled	Enrolled Outside of the Municipality
	(1)	(2)	(3)	(4)
Treated	0.025***	0.018***	0.007***	0.002*
	(0.007)	(0.005)	(0.002)	(0.001)
Observations	17,704	17,704	17,704	17,704
Pre-policy Mean	0.18	0.01	0.01	0.00

Note: Standard errors in parentheses, clustered at the municipality level. All specifications include municipality-level control variables, measured pre-policy and discretized. ATTs are calculated using inverse probability weighting (IPW) and agregated to generate an overall average effect following Callaway and Sant'Anna (2021). Observations vary between outcomes due to years of data availability. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Figure 8: Likelihood of Enrolling in a Federal HEI



7.2.2 Tertiary education costs

Since public higher education institutions in Brazil are free, costs associated with higher education can be thought of as a function of proximity to a public HEI. By reducing the distance to a public HEI, this policy directly lowers the costs of attending higher education for treated students, who will no longer incur moving or significant transportation costs. As shown in the theoretical framework section, if the policy reduces the cost of attending higher education, the share of students graduating high school should increase, all else constant. Although I cannot empirically measure the role of costs, it is reasonable to assume the policy is reducing costs related to higher-education attendance by reducing distance.

7.3 The role of returns to education

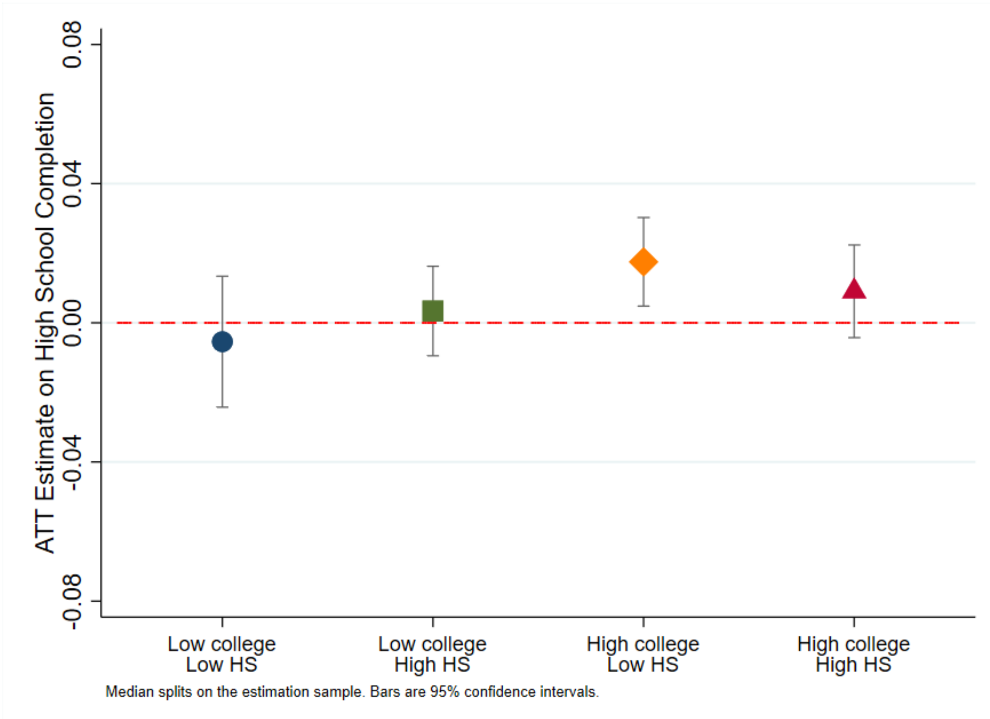
As derived within the framework, the relative magnitudes of local college and high school premiums determine whether an expansion in higher-education opportunities can affect the high school decision. One condition is that at least some students need to choose not to graduate high school in the absence of higher education. That implies that high school wages cannot exceed primary education earnings to the point that high school would be strictly preferred over dropping out for all students. The larger the high school wages in a municipality, the smaller the pool of students at (no high school, no college) that can be moved by the policy.

Additionally, for transitions from (no high school, no college) to (high school, college) to occur, college premiums need to be sufficiently large to drive students to graduate from high school *because* of new college opportunities. From these two conditions, it follows that changes in the high school decision should occur only where college premiums are high, especially when they are high *relative to* high school premiums. In these places, there will be a larger pool of students who do not graduate from high school, and sizeable college premiums play a key role in increasing the expected returns to high school.

Using pre-policy data from the 2000 Brazilian Census, I construct municipality-level pre-policy college and high school premiums, defined as $\log(\frac{y_m^C}{y_m^{HS}})$ and $\log(\frac{y_m^{HS}}{y_m^{NH}})$, respectively, where y_m^C, y_m^{HS} and y_m^{NH} are municipality level average earnings of workers with higher education, secondary school and primary school degrees. I then divide municipalities into four groups: low college premium and low high school premium; low college premium and high high school premium; high college premium and low high school premium; and high college premium and high high school premium.

Low and high premiums are defined as below and above the median, respectively. Figure 9 plots the average treatment effects of receiving an HEI on high school graduation rates for each of these groups.

Figure 9: High School Completion effect, by pre-policy distribution of college and high school premium



Although the estimates are not statistically different, the point estimates are largest for places with a high college premium pre-policy. Effects are also largest in municipalities where college premium was high and high school premium was low. This is in line with the model's predictions that college earnings would play a larger role in places where high school earnings alone were not enough to drive students to complete high school pre-policy.

This result suggests that the overall effects shown in Table 3 mask important heterogeneity in who is responding to increased tertiary educational opportunities by changing pre-tertiary schooling decisions. This is also suggestive evidence that the key channel through which future educational opportunities affect earlier educational attainment is by increasing expected returns to schooling, which operates through the increased probability of accessing college-level wages. This is especially relevant in settings where returns to high school are modest relative to returns to higher education, which is precisely the case in Brazil and other LMICs.

8 How sustainable is the policy effect in the long run?

As students respond to this policy by increasing high school and college attainment, returns to both levels of education might decrease, diminishing the impact of higher education opportunities on high school completion over time. In this section, I provide suggestive evidence on how municipality-level college and high school premiums (in the formal labor market) behave following the opening of an HEI. This will inform on whether we can expect higher education expansion to permanently increase the returns to high school, or whether these short-run effects should dissipate over time.

Figure 10: Municipality college-premium (left) and high-school premium (right) in the formal labor market

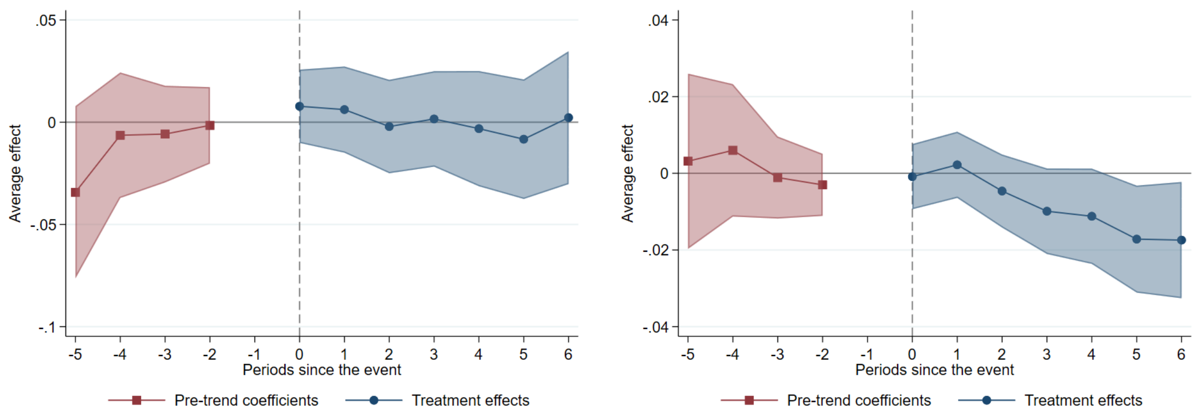


Figure 11: Primary education wages (left) and secondary education wages (right) in the formal labor market

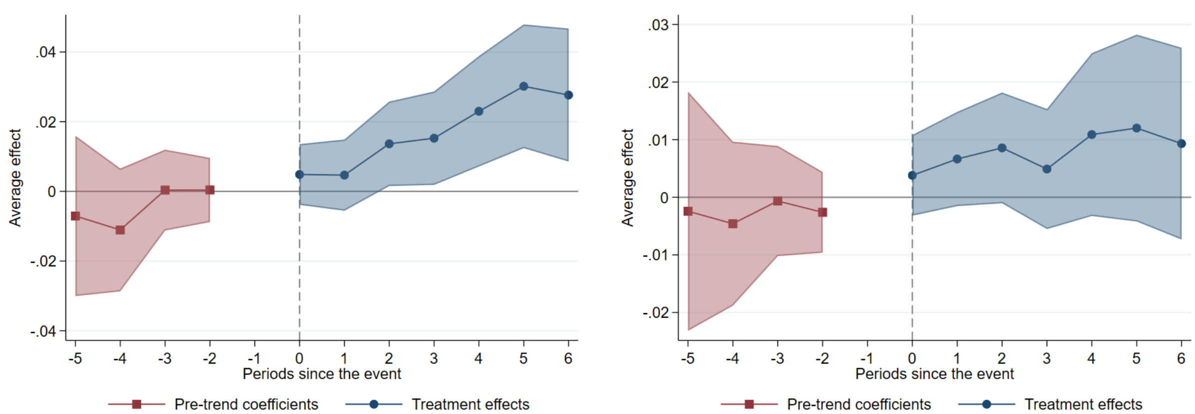


Figure 10 shows that local college premiums remain stable for up to 6 years after an HEI's opening. On the other hand, high-school premiums begin to decrease sharply two years after the policy. This decrease in high school returns comes from an increase in primary-education wages, rather than a decrease in earnings for high school graduates, as shown in Figure 11. This is likely a result of both a decreased supply of primary-educated young workers due to increased educational attainment following the opening of an HEI, and the potential creation of new jobs that might increase demand for low-skilled labor. Overall, college premiums seem to be constant 6 years after the policy, suggesting that the higher-education-induced increase in returns to high school can be sustained over time, provided that college returns remain high relative to the outside option (in this case, primary education).

9 Robustness analysis

Table 7: Effects of Higher Education Expansion on Student Composition

Panel A: *5th Grade*			
	White	Studied in Public Schools Only	Mother Completed High School
	(1)	(2)	(3)
Treated	-0.008*	-0.012	0.000
	(0.004)	(0.007)	(0.004)
Observations	14,470	14,471	14,471
Pre-policy Mean	0.35	0.57	0.29
Panel B: *9th Grade*			
	White	Studied in Public Schools Only	Mother Completed High School
	(1)	(2)	(3)
Treated	0.002	0.020***	-0.001
	(0.003)	(0.005)	(0.004)
Observations	15,093	15,093	15,092
Pre-policy Mean	0.35	0.90	0.31

Note: Standard errors in parentheses, clustered at the municipality level. All specifications include municipality-level control variables, measured pre-policy and discretized. ATTs are calculated using inverse probability weighting (IPW) and aggregated to generate an overall average effect following Callaway and Sant'Anna (2021). Observations vary between outcomes due to years of data availability. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

One possible threat to identification is that, due to a potential influx of people into the municipality caused by the arrival of an HEI, such as professors, staff, and new students, the composition of public K-12 schools could change in ways that affect high school graduation rates and test scores. The observed results would then not be attributable to the expansion of higher education itself, but to compositional changes generated by the policy. To assess whether this is the case, I replace the outcome variable in my estimation with the shares of white students, students who attended only public schools throughout their lives, and students whose mothers completed high school. This will measure whether the policy affects student composition[9].

As shown in Table 7, student composition does not appear to change as a result of the policy. In fact, all significant estimates would bias the results downward, given that non-white and public school students, on average, have lower high school completion rates and lower test scores than white and private school students.

10 Discussion

The key role of future educational opportunities in shaping early educational attainment has only recently begun to be explored in the literature and has not yet reached the policy debate as a valuable tool for sustainably increasing demand for basic schooling. By providing low-income students with the same higher-education opportunities as high-income students, expansion policies can meaningfully increase pre-tertiary educational attainment and reduce disparities. Exploring a large-scale expansion of higher education institutions in Brazil, I find that local students exposed to a new HEI are 2% more likely to graduate high school. More disadvantaged, non-white public students benefit the most: they are 3% more likely to enroll in high school following the opening of an HEI (which represents 1/3 of the pre-policy high school enrollment gap relative to white students) and, conditional on doing so, are 4% more likely to graduate high school.

Trade schools and universities seem to play similar roles in shaping schooling decisions. However, these two types of future schooling opportunities seem to impact students at different stages of their educational career. 9th graders who reside in municipalities that received a university were 2 percentage points more likely to transition to high school, with no impact on those residing in municipalities that received a trade school. Still, trade schools seem to shift schooling decisions of older pre-tertiary students who would have started high school in the absence of the policy. This result is particularly relevant for LMICs, since trade schools are smaller and less costly than four-year universities, and potentially more aligned with the needs of the local labor markets.

My results stem from increased educational opportunities rather than from indirect labor market effects. I find that students are 14% more likely to apply to an HEI and experience a nearly 1-percentage-point increase in the likelihood of enrolling (a 70% increase from the pre-policy mean).

[9]Public school attendance in Brazil is highly correlated with income. The questionnaire asks students if, starting in first grade, they studied in public only, private only, or both private and public schools.

On the other hand, changes in local wage levels through newly created jobs or general equilibrium effects are not prevalent, as the changes I observe in earnings for primary-educated young workers would push the effects in the opposite direction by making dropping out more attractive.

A key prediction of the theoretical framework is that higher education opportunities should only matter if college premiums are high relative to high school returns. I find empirical evidence to support this prediction. Only in municipalities with a high college premium pre-policy do students respond to a new HEI by increasing secondary schooling completion. This effect is largest in municipalities where the college premium is high, and the high school premium is low, highlighting the important role that future educational opportunities play in making pre-tertiary education more worthwhile.

Last, I examine whether these results can be sustainable over time by analyzing how the college and high school premium in the formal labor market change up to six years after the opening of an HEI. While college premiums remain stable over time, high school premiums decline sharply as primary-level wages rise. This is consistent with the general equilibrium wage effects documented by Khanna (2023), who finds that expanding primary education in India raised wages for unskilled workers.

My empirical results contrast with Bedard (2001). While she finds increases in dropout rates resulting from the expansion of higher education in the United States, I find that students exposed to new HEIs responded by increasing their high school completion rates. This difference, however, is likely due to contextual differences, and both results are supported by my framework. The average years of schooling in the United States in the 1970s was approximately 12 years, equivalent to high school completion. In the context of this paper, the average educational attainment is below 9 years, which is equivalent to primary school completion in Brazil. A necessary condition for an expansion in higher education to affect high school decisions is that a sufficient number of students are dropping out of high school before the policy is implemented and thus will be able to change their schooling decisions after the opening of an HEI. This is true in my setting, but less so in the United States. In addition, the signaling model in Bedard (2001) proposes that, following the opening of an HEI, only high-type students would attend college and low-type students, for whom college attendance is too costly, would not be able to make it past secondary schooling. This would lead to a separating equilibrium and allow employers to correctly identify students' types, paying low wages to low-types and high wages to high-type students. In my framework, if low-type students were to continue graduating from high school, a separating equilibrium would be equivalent to the policy decreasing high school-level wages, which is predicted to reduce high school completion and increase dropouts, as found by Bedard (2001).

In a context more closely related to this paper, I also build on previous work by Jagnani and Khanna (2020) and provide further external validity for their findings. Importantly, I find significant gains in persistence at lower levels of education following higher-education expansion in a setting where most new HEIs are newly created, not prestigious, and predominantly technically focused (trade schools). I show that not only four-year colleges, but also smaller two-year trade schools can have important trickle-down effects on high school completion. By developing a theoretical framework and conducting an empirical analysis of the mechanisms, I explore the channels through which these effects may operate and highlight the key role of skill premium and increased college access opportunities in shaping this relationship.

Overall, the expansion of higher education in Brazil had an important effect in shifting earlier educational decisions for pre-tertiary students in receiving municipalities, generating meaningful impacts on students' persistence and graduation rates. Students were also more likely to attend federal higher education institutions. This policy provides a relevant example of how the availability of future educational opportunities, coupled with an effective increase in access for low-income students, can increase returns to basic education and generate meaningful improvements in overall human capital accumulation.

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A Complete overview of the model

A.1 Setup

At the end of 9th grade, a student chooses a schooling path $(s, d) \in \{(0,0),(1,0),(1,1)\}$, where $s = 1$ denotes graduating high school and $d = 1$ denotes attempting college. Because college requires first graduating high school, the pair $(0, 1)$ is not feasible. The student maximizes expected utility:

$$\max_{(s,d)\in\{(0,0),(1,0),(1,1)\}} U(C_1) + \beta_2\,\mathbb{E}\big[U(C_2)\big] + \beta_3\,\mathbb{E}\big[U(C_3)\big], \tag{25}$$

where $U(c) = \ln c$ and the expectation is over the admission lottery. For paths $(0, 0)$ and $(1, 0)$ there is no uncertainty, so the expectation is trivial. For path $(1, 1)$, the student is admitted with probability $\pi_c \in (0, 1)$; conditional on admission they enroll and pay costs p^c, and conditional on non-admission they earn high-school wages. When choosing s and d, a student knows their high-school cost, p^H, college cost, p^c, and the likelihood of being accepted to an HEI, π_c, but does not know the college admission outcome.

Period 1. Period 1 lasts 3 years (high school duration in Brazil, grades 10–12):

$$C_1 = \bar{k}_1 + (1 - s)\, y_1^{NH} - s\, p^H,$$

where $\bar{k}_1 = 3\bar{k}$ and $y_1^{NH} = 3y^{NH}$.

Period 2. Period 2 lasts 4 years (the typical higher education duration), with $\bar{k}_2 = 4\bar{k}$:

$$C_2 = \begin{cases} \bar{k}_2 + y_2^{NH} & \text{if } (s,d) = (0,0), \\ \bar{k}_2 + y_2^{HS} & \text{if } (s,d) = (1,0), \\ \bar{k}_2 + y_2^{HS} & \text{if } (s,d) = (1,1) \text{ and not admitted (probability } 1-\pi_c), \\ \bar{k}_2 - p^c & \text{if } (s,d) = (1,1) \text{ and admitted (probability } \pi_c), \end{cases}$$

where $y_2^{NH} = 4y^{NH}$ and $y_2^{HS} = 4y^{HS}$.

Period 3. Period 3 represents the remainder of working life. C_3 denotes *per-year* consumption in that period, which is assumed to be constant:

$$C_3 = \begin{cases} \bar{k} + y^{NH} & \text{if } (s,d) = (0,0), \\ \bar{k} + y^{HS} & \text{if } (s,d) = (1,0), \\ \bar{k} + y^{HS} & \text{if } (s,d) = (1,1) \text{ and not admitted}, \\ \bar{k} + y^{C} & \text{if } (s,d) = (1,1) \text{ and admitted}. \end{cases}$$

Timing and discount factors. Period 1 lasts 3 years, period 2 lasts 4 years, and period 3 runs from year 8 to the end of working life, year N. Accordingly,

$$\beta_2 = \beta^3, \qquad \beta_3 \equiv \sum_{t=8}^{N} \beta^{t-1} = \beta^7 \frac{1-\beta^{N-7}}{1-\beta} > 0. \tag{26}$$

Because C_3 is measured per year, β_3 is the exact present value, as of period 1, of one unit of per-year consumption received in every year $t = 8, \ldots, N$.

Wages. y^{NH} is the wage for a primary-educated worker, y^{HS} for a high-school graduate, and y^C for a college graduate. I assume all are constant over time. An individual's education level is fixed after the schooling decision is made and the college admission lottery is realized.

Endowment and costs. $\bar{k}$ is the household endowment per year. $p^H > 0$ is the direct cost of attending high school and $p^c > 0$ is the direct cost of attending college. Both p^H and p^c do not include tuition costs, and are mostly comprised of transportation or relocation costs. Throughout, I assume $p^H < \bar{k}_1$ and $p^c < \bar{k}_2$ so that period-1 and period-2 consumption are strictly positive on all paths.

College access. $\pi_c \in (0,1)$ is the probability of admission to a higher-education institution. The student knows π_c when choosing (s,d), but the admission outcome is only realized after the choice.

Cost distributions. Costs are drawn independently and uniformly: $p^H \sim U[0, \bar{P}^H]$ and $p^c \sim U[0, \bar{P}^c]$.

A.2 Value Functions

Substituting $U(c) = \ln c$ and taking the expectation over the admission lottery for path $(1,1)$:

$$V(0,0) = \ln(\bar{k}_1 + y_1^{NH}) + \beta_2 \ln(\bar{k}_2 + y_2^{NH}) + \beta_3 \ln(\bar{k} + y^{NH}), \tag{27}$$

$$V(1,0) = \ln(\bar{k}_1 - p^H) + \beta_2 \ln(\bar{k}_2 + y_2^{HS}) + \beta_3 \ln(\bar{k} + y^{HS}), \tag{28}$$

$$\begin{aligned} V(1,1) = {} & \ln(\bar{k}_1 - p^H) + \beta_2 \big[(1-\pi_c)\ln(\bar{k}_2 + y_2^{HS}) + \pi_c \ln(\bar{k}_2 - p^c)\big] \\ & + \beta_3 \big[(1-\pi_c)\ln(\bar{k} + y^{HS}) + \pi_c \ln(\bar{k} + y^C)\big]. \end{aligned} \tag{29}$$

A.3 Decision Thresholds and Decision Rule

A.3.1 Threshold 1: $(1,0)$ vs. $(0,0)$

Derivation. A student chooses $(1,0)$ over $(0,0)$ when $V(1,0) \geq V(0,0)$. Setting $V(1,0) = V(0,0)$ using (28) and (27):

$$\ln(\bar{k}_1 - p^H) + \beta_2 \ln(\bar{k}_2 + y_2^{HS}) + \beta_3 \ln(\bar{k} + y^{HS}) = \ln(\bar{k}_1 + y_1^{NH}) + \beta_2 \ln(\bar{k}_2 + y_2^{NH}) + \beta_3 \ln(\bar{k} + y^{NH}).$$

Isolating p^H on one side, I can find a threshold for which a student will be indifferent between $(1,0)$ and $(0,0)$:

$$\begin{aligned} \ln(\bar{k}_1 - p^H) - \ln(\bar{k}_1 + y_1^{NH}) &= \beta_2 \big[\ln(\bar{k}_2 + y_2^{NH}) - \ln(\bar{k}_2 + y_2^{HS})\big] + \beta_3 \big[\ln(\bar{k} + y^{NH}) - \ln(\bar{k} + y^{HS})\big] \\ \implies \quad \ln\left(\frac{\bar{k}_1 - p^H}{\bar{k}_1 + y_1^{NH}}\right) &= \beta_2 \ln\left(\frac{\bar{k}_2 + y_2^{NH}}{\bar{k}_2 + y_2^{HS}}\right) + \beta_3 \ln\left(\frac{\bar{k} + y^{NH}}{\bar{k} + y^{HS}}\right) \\ \implies \quad \ln\left(\frac{\bar{k}_1 - p^H}{\bar{k}_1 + y_1^{NH}}\right) &= \ln\left[\left(\frac{\bar{k}_2 + y_2^{NH}}{\bar{k}_2 + y_2^{HS}}\right)^{\beta_2} \left(\frac{\bar{k} + y^{NH}}{\bar{k} + y^{HS}}\right)^{\beta_3}\right] \\ \implies \quad \bar{p}_{10}^H &= \bar{k}_1 - (\bar{k}_1 + y_1^{NH}) \left(\frac{\bar{k}_2 + y_2^{NH}}{\bar{k}_2 + y_2^{HS}}\right)^{\beta_2} \left(\frac{\bar{k} + y^{NH}}{\bar{k} + y^{HS}}\right)^{\beta_3}. \end{aligned}$$

Since $V(1,0) - V(0,0)$ is strictly decreasing in p^H, a student chooses $(1,0)$ over $(0,0)$ if and only if $p^H \leq \bar{p}_{10}^H$.

A.3.2 Threshold 2: $(1,1)$ vs. $(1,0)$

Derivation. A student chooses $(1,1)$ over $(1,0)$ when $V(1,1) \geq V(1,0)$. Setting $V(1,1) = V(1,0)$. The terms $\ln(\bar{k}_1 - p^H)$ are identical in (29) and (28) and cancel, leaving

$$\begin{aligned}\beta_2\big[(1-\pi_c)\ln(\bar{k}_2 + y_2^{HS}) + \pi_c\ln(\bar{k}_2 - p^c)\big] + \beta_3\big[(1-\pi_c)\ln(\bar{k} + y^{HS}) + \pi_c\ln(\bar{k} + y^C)\big] \\ = \beta_2\ln(\bar{k}_2 + y_2^{HS}) + \beta_3\ln(\bar{k} + y^{HS}).\end{aligned}$$

Subtract the right side from the left. In the β_2 block, $(1-\pi_c)\ln(\bar{k}_2 + y_2^{HS}) - \ln(\bar{k}_2 + y_2^{HS}) = -\pi_c\ln(\bar{k}_2 + y_2^{HS})$, and similarly in the β_3 block:

$$\pi_c\beta_2\big[\ln(\bar{k}_2 - p^c) - \ln(\bar{k}_2 + y_2^{HS})\big] + \pi_c\beta_3\big[\ln(\bar{k} + y^C) - \ln(\bar{k} + y^{HS})\big] = 0.$$

Dividing through by $\pi_c > 0$ and $\beta_2 > 0$, and exponentiating:

$$\frac{\bar{k}_2 - p^c}{\bar{k}_2 + y_2^{HS}} = \left(\frac{\bar{k} + y^{HS}}{\bar{k} + y^C}\right)^{\beta_3/\beta_2}.$$

Multiplying by $(\bar{k}_2 + y_2^{HS}) > 0$ and subtracting from $\bar{k}_2$:

$$\bar{p}^c_{11|10} = \bar{k}_2 - (\bar{k}_2 + y_2^{HS})\left(\frac{\bar{k} + y^{HS}}{\bar{k} + y^C}\right)^{\beta_3/\beta_2}. \tag{30}$$

Since $V(1,1) - V(1,0)$ is strictly decreasing in p^c, a student chooses $(1,1)$ over $(1,0)$ if and only if $p^c \leq \bar{p}^c_{11|10}$. Additionally, since $(1,0)$ is available for that student, it must be true that $p^H \leq \bar{p}^H_{10}$.

A.3.3 Threshold 3: $(1,1)$ vs. $(0,0)$

Derivation. A student chooses $(1,1)$ over $(0,0)$ when $V(1,1) \geq V(0,0)$. Setting $V(1,1) = V(0,0)$:

$$\begin{aligned}\ln(\bar{k}_1 - p^H) + \beta_2\big[(1-\pi_c)\ln(\bar{k}_2 + y_2^{HS}) + \pi_c\ln(\bar{k}_2 - p^c)\big] + \beta_3\big[(1-\pi_c)\ln(\bar{k} + y^{HS}) + \pi_c\ln(\bar{k} + y^C)\big] \\ = \ln(\bar{k}_1 + y_1^{NH}) + \beta_2\ln(\bar{k}_2 + y_2^{NH}) + \beta_3\ln(\bar{k} + y^{NH}).\end{aligned}$$

Isolating p^c:

$$\pi_c \beta_2 \ln(\bar{k}_2 - p^c) = \ln\left(\frac{\bar{k}_1 + y_1^{NH}}{\bar{k}_1 - p^H}\right) + \beta_2 \ln(\bar{k}_2 + y_2^{NH}) - \beta_2(1-\pi_c)\ln(\bar{k}_2 + y_2^{HS}) \\ + \beta_3 \ln(\bar{k} + y^{NH}) - \beta_3(1-\pi_c)\ln(\bar{k} + y^{HS}) - \beta_3 \pi_c \ln(\bar{k} + y^C).$$

Divide every term by $\pi_c\beta_2 > 0$ and rearranging:

$$\ln(\bar{k}_2 - p^c) = \frac{1}{\pi_c}\left\{\frac{1}{\beta_2}\ln\left(\frac{\bar{k}_1 + y_1^{NH}}{\bar{k}_1 - p^H}\right) + \ln\left(\frac{\bar{k}_2 + y_2^{NH}}{(\bar{k}_2 + y_2^{HS})^{1-\pi_c}}\right) + \frac{\beta_3}{\beta_2}\ln\left(\frac{\bar{k} + y^{NH}}{(\bar{k} + y^{HS})^{1-\pi_c}(\bar{k} + y^C)^{\pi_c}}\right)\right\}.$$

$$\bar{p}^c_{11|00}(p^H) = \bar{k}_2 - \left[E(p^H)\right]^{1/\pi_c}, \tag{31}$$

where, written out in full,

$$E(p^H) \equiv \left(\frac{\bar{k}_1 + y_1^{NH}}{\bar{k}_1 - p^H}\right)^{1/\beta_2} \cdot \frac{\bar{k}_2 + y_2^{NH}}{(\bar{k}_2 + y_2^{HS})^{1-\pi_c}} \cdot \left(\frac{\bar{k} + y^{NH}}{(\bar{k} + y^{HS})^{1-\pi_c}(\bar{k} + y^C)^{\pi_c}}\right)^{\beta_3/\beta_2} > 0. \tag{32}$$

By the same monotonicity argument as in Section A.3.2, a student chooses $(1,1)$ over $(0,0)$ if and only if $p^c \le \bar{p}^c_{11|00}(p^H)$. Since a student is comparing $(1,1)$ to $(0,0)$ and not $(1,0)$, it must be true that $p^H > \bar{p}^H_{10}$ and $(1,0)$ is not available for that student. Note that $\bar{p}^c_{11|00}$ is a *function* of p^H, since p^H appears in $E(p^H)$.

Boundary condition $\bar{p}^c_{11|00}(\bar{p}^H_{10}) = \bar{p}^c_{11|10}$. Evaluating $\bar{p}^c_{11|00}$ at $p^H = \bar{p}^H_{10}$ and using

$$\Pi \equiv \left(\frac{\bar{k}_2 + y_2^{NH}}{\bar{k}_2 + y_2^{HS}}\right)^{\beta_2}\left(\frac{\bar{k} + y^{NH}}{\bar{k} + y^{HS}}\right)^{\beta_3} > 0, \qquad \text{so} \qquad \bar{p}^H_{10} = \bar{k}_1 - (\bar{k}_1 + y_1^{NH})\,\Pi.$$

we have that $\bar{k}_1 - \bar{p}^H_{10} = (\bar{k}_1 + y_1^{NH})\Pi$, so:

$$\left(\frac{\bar{k}_1 + y_1^{NH}}{\bar{k}_1 - \bar{p}^H_{10}}\right)^{1/\beta_2} = \Pi^{-1/\beta_2} = \frac{\bar{k}_2 + y_2^{HS}}{\bar{k}_2 + y_2^{NH}}\left(\frac{\bar{k} + y^{HS}}{\bar{k} + y^{NH}}\right)^{\beta_3/\beta_2}.$$

Substituting into (32), the period-2 factors give

$$\frac{\bar{k}_2 + y_2^{HS}}{\bar{k}_2 + y_2^{NH}} \cdot \frac{\bar{k}_2 + y_2^{NH}}{(\bar{k}_2 + y_2^{HS})^{1-\pi_c}} = (\bar{k}_2 + y_2^{HS})^{\pi_c},$$

since $(\bar{k}_2 + y_2^{NH})$ cancels and $x/x^{1-\pi_c} = x^{\pi_c}$; and the period-3 factors give

$$\left(\frac{\bar{k}+y^{HS}}{\bar{k}+y^{NH}} \cdot \frac{\bar{k}+y^{NH}}{(\bar{k}+y^{HS})^{1-\pi_c}(\bar{k}+y^{C})^{\pi_c}}\right)^{\beta_3/\beta_2} = \left(\frac{\bar{k}+y^{HS}}{\bar{k}+y^{C}}\right)^{\pi_c\beta_3/\beta_2}.$$

Hence $E(\bar{p}^H_{10}) = (\bar{k}_2+y_2^{HS})^{\pi_c}\big((\bar{k}+y^{HS})/(\bar{k}+y^{C})\big)^{\pi_c\beta_3/\beta_2}$, and raising to $1/\pi_c$ divides every exponent by π_c:

$$\bar{k}_2 - \bar{p}^c_{11|00}(\bar{p}^H_{10}) = (\bar{k}_2 + y_2^{HS})\left(\frac{\bar{k}+y^{HS}}{\bar{k}+y^{C}}\right)^{\beta_3/\beta_2} = \bar{k}_2 - \bar{p}^c_{11|10}.$$

Therefore

$$\bar{p}^c_{11|00}(\bar{p}^H_{10}) = \bar{p}^c_{11|10}. \tag{33}$$

This is the internal consistency condition of the model: at $p^H = \bar{p}^H_{10}$ the student is indifferent between $(1,0)$ and $(0,0)$, so comparing $(1,1)$ against either must yield the same critical college cost.

Strict monotonicity of $\bar{p}^c_{11|00}(\cdot)$. Writing $\big((\bar{k}_1 + y_1^{NH})/(\bar{k}_1 - p^H)\big)^{1/\beta_2} = (\bar{k}_1 + y_1^{NH})^{1/\beta_2}(\bar{k}_1 - p^H)^{-1/\beta_2}$ and raising $E(p^H)$ to $1/\pi_c$, the p^H-dependent factor becomes $(\bar{k}_1 - p^H)^{-1/(\pi_c\beta_2)}$. Setting $\alpha \equiv 1/(\pi_c\beta_2) > 0$,

$$\bar{p}^c_{11|00}(p^H) = \bar{k}_2 - \Lambda\,(\bar{k}_1 - p^H)^{-\alpha}, \Lambda \equiv \left[\frac{(\bar{k}_1 + y_1^{NH})^{1/\beta_2}(\bar{k}_2 + y_2^{NH})}{(\bar{k}_2 + y_2^{HS})^{1-\pi_c}}\left(\frac{\bar{k}+y^{NH}}{(\bar{k}+y^{HS})^{1-\pi_c}(\bar{k}+y^{C})^{\pi_c}}\right)^{\beta_3/\beta_2}\right]^{1/\pi_c} > 0, \tag{34}$$

where Λ contains no p^H. Differentiating:

$$\frac{\partial \bar{p}^c_{11|00}(p^H)}{\partial p^H} = -\Lambda\,\alpha\,(\bar{k}_1 - p^H)^{-\alpha-1} < 0.$$

So $\bar{p}^c_{11|00}(\cdot)$ is strictly decreasing, and with (33),

$$\bar{p}^c_{11|00}(p^H) < \bar{p}^c_{11|00}(\bar{p}^H_{10}) = \bar{p}^c_{11|10} \qquad \text{for every } p^H > \bar{p}^H_{10}. \tag{35}$$

A student who has paid a larger p^H has lower period-1 consumption, so the critical college cost at which she is willing to take the college path is lower. A useful identity, immediate from (34), is

$$\Lambda\,(\bar{k}_1 - p^H)^{-\alpha} = \bar{k}_2 - \bar{p}^c_{11|00}(p^H), \qquad \text{and in particular} \qquad \Lambda\,(\bar{k}_1 - \bar{p}^H_{10})^{-\alpha} = \bar{k}_2 - \bar{p}^c_{11|10}. \tag{36}$$

A.3.4 Decision Rule

Combining the three thresholds:

$$(s,d)^* = \begin{cases} (1,1) & \text{if } \big[p^H \le \bar{p}^H_{10} \text{ and } p^c \le \bar{p}^c_{11|10}\big] \text{ or } \big[p^H > \bar{p}^H_{10} \text{ and } p^c \le \bar{p}^c_{11|00}(p^H)\big], \\ (1,0) & \text{if } p^H \le \bar{p}^H_{10} \text{ and } p^c > \bar{p}^c_{11|10}, \\ (0,0) & \text{if } p^H > \bar{p}^H_{10} \text{ and } p^c > \bar{p}^c_{11|00}(p^H). \end{cases} \tag{37}$$

The rule is exhaustive and the four cases are internally consistent:

- $p^H \le \bar{p}^H_{10}$, $p^c \le \bar{p}^c_{11|10}$: Threshold 1 gives $(1,0) \succ (0,0)$ and Threshold 2 gives $(1,1) \succ (1,0)$, so by transitivity $(1,1)$ is chosen.
- $p^H \le \bar{p}^H_{10}$, $p^c > \bar{p}^c_{11|10}$: $(1,0) \succ (0,0)$ and $(1,0) \succ (1,1)$, so $(1,0)$ is chosen.
- $p^H > \bar{p}^H_{10}$, $p^c \le \bar{p}^c_{11|00}(p^H)$: $(0,0) \succ (1,0)$, so $(1,0)$ is dominated; Threshold 3 gives $(1,1) \succ (0,0)$ and transitivity gives $(1,1) \succ (1,0)$, so $(1,1)$ is chosen.
- $p^H > \bar{p}^H_{10}$, $p^c > \bar{p}^c_{11|00}(p^H)$: $(0,0)$ dominates both alternatives.

A.4 Aggregate High School Graduation

Definition. S^* is the aggregate share of students who graduate high school, i.e. who choose $s = 1$ either through $(1,0)$ or $(1,1)$.

Derivation. From (37), a student graduates in exactly one of two mutually exclusive ways: either $p^H \le \bar{p}^H_{10}$, in which case she chooses $(1,0)$ or $(1,1)$ depending on p^c but graduates either way, or $p^H > \bar{p}^H_{10}$ and $p^c \le \bar{p}^c_{11|00}(p^H)$, in which case she moves directly from $(0,0)$ to $(1,1)$. Hence

$$S^* = \Pr(p^H \le \bar{p}^H_{10}) + \Pr\big(p^H > \bar{p}^H_{10} \text{ and } p^c \le \bar{p}^c_{11|00}(p^H)\big).$$

Since the threshold $\bar{p}^c_{11|00}(p^H)$ depends on the realization of p^H, I apply the law of total probability and conditioning on p^H to get:

$$\Pr\big(p^H > \bar{p}^H_{10} \text{ and } p^c \le \bar{p}^c_{11|00}(p^H)\big) = \int_{\bar{p}^H_{10}}^{\bar{P}^H} \Pr\big(p^c \le \bar{p}^c_{11|00}(p^H) \,\big|\, p^H\big)\, f(p^H)\, dp^H,$$

with $f(p^H) = 1/\bar{P}^H$. Because the two cost draws are *independent random variables*, the conditional probability equals the unconditional distribution function of p^c evaluated at the realized threshold: $\Pr(p^c \le \bar{p}^c_{11|00}(p^H) \mid p^H) = \bar{p}^c_{11|00}(p^H)/\bar{P}^c$, valid whenever $\bar{p}^c_{11|00}(p^H) \in [0, \bar{P}^c]$, which I assume throughout (Assumption 2 below), together with $\bar{p}^H_{10} \in (0, \bar{P}^H)$. Writing $\theta_H \equiv \bar{p}^H_{10}/\bar{P}^H$,

$$S^* = \theta_H + \frac{1}{\bar{P}^H \bar{P}^c} \int_{\bar{p}^H_{10}}^{\bar{P}^H} \bar{p}^c_{11|00}(p^H)\, dp^H. \tag{38}$$

[Regularity] $\bar{p}^H_{10} \in (0, \bar{P}^H)$.

[Interior solution] $\bar{p}^c_{11|00}(p^H) \in (0, \bar{P}^c)$ for all $p^H \in [\bar{p}^H_{10}, \bar{P}^H]$.

Closed form. Substituting $\bar{p}^c_{11|00}(p^H)$ below into (38)

$$\bar{p}^c_{11|00}(p^H) = \bar{k}_2 - \Lambda\, (\bar{k}_1 - p^H)^{-\alpha}, \Lambda \equiv \left[\frac{(\bar{k}_1 + y_1^{NH})^{1/\beta_2} (\bar{k}_2 + y_2^{NH})}{(\bar{k}_2 + y_2^{HS})^{1-\pi_c}} \left(\frac{\bar{k} + y^{NH}}{(\bar{k} + y^{HS})^{1-\pi_c} (\bar{k} + y^C)^{\pi_c}} \right)^{\beta_3/\beta_2} \right]^{1/\pi_c} > 0,$$

We have:

$$\int_{\bar{p}^H_{10}}^{\bar{P}^H} \bar{p}^c_{11|00}(p^H)\, dp^H = \bar{k}_2(\bar{P}^H - \bar{p}^H_{10}) - \Lambda \int_{\bar{p}^H_{10}}^{\bar{P}^H} (\bar{k}_1 - p^H)^{-\alpha}\, dp^H,$$

and substituting $u = \bar{k}_1 - p^H$, $du = -dp^H$,

$$\int_{\bar{p}^H_{10}}^{\bar{P}^H} (\bar{k}_1 - p^H)^{-\alpha}\, dp^H = \int_{\bar{k}_1 - \bar{P}^H}^{\bar{k}_1 - \bar{p}^H_{10}} u^{-\alpha}\, du = \frac{(\bar{k}_1 - \bar{p}^H_{10})^{1-\alpha} - (\bar{k}_1 - \bar{P}^H)^{1-\alpha}}{1-\alpha},$$

valid for $\alpha \neq 1$, i.e. $\pi_c \beta_2 \neq 1$. The expression is strictly positive in both cases: for $\alpha < 1$ the numerator is positive and $1 - \alpha > 0$; for $\alpha > 1$ both are negative. Hence

$$S^* = \theta_H + \frac{1}{\bar{P}^H \bar{P}^c} \underbrace{\left\{ \bar{k}_2(\bar{P}^H - \bar{p}^H_{10}) - \Lambda\, \frac{(\bar{k}_1 - \bar{p}^H_{10})^{1-\alpha} - (\bar{k}_1 - \bar{P}^H)^{1-\alpha}}{1-\alpha} \right\}}_{\equiv\ \mathcal{S}\ =\ \int_{\bar{p}^H_{10}}^{\bar{P}^H} \bar{p}^c_{11|00}(p^H)\, dp^H} \tag{39}$$

A.4.1 Monotonicity of S^*

If the share of individuals completing high school as represented by S^* is a monotonic function of the thresholds, I can analyze changes in the thresholds to understand how policy-induced changes in parameters can affect S^*.

Independence from $\bar{p}^c_{11|10}$. $\bar{p}^c_{11|10}$ governs only the split between $(1,0)$ and $(1,1)$ among students with $p^H \leq \bar{p}^H_{10}$. Both choices involve graduating, so moving a student between them leaves S^* unchanged. Therefore, $\bar{p}^c_{11|10}$ appears in neither (38) nor (39).

S^* is strictly increasing in $\bar{p}^H_{10}$. Differentiating (39) directly: $\partial\theta_H/\partial\bar{p}^H_{10} = 1/\bar{P}^H$ and $\partial\big[\bar{k}_2(\bar{P}^H - \bar{p}^H_{10})\big]/\partial\bar{p}^H_{10} = -\bar{k}_2$. Since only $(\bar{k}_1 - \bar{p}^H_{10})^{1-\alpha}$ depends on $\bar{p}^H_{10}$ in the remaining term, and by the chain rule, we have that $\partial(\bar{k}_1 - \bar{p}^H_{10})^{1-\alpha}/\partial\bar{p}^H_{10} = -(1-\alpha)(\bar{k}_1 - \bar{p}^H_{10})^{-\alpha}$, then:

$$\frac{\partial}{\partial\bar{p}^H_{10}}\left[-\Lambda\,\frac{(\bar{k}_1 - \bar{p}^H_{10})^{1-\alpha} - (\bar{k}_1 - \bar{P}^H)^{1-\alpha}}{1-\alpha}\right] = +\Lambda\,(\bar{k}_1 - \bar{p}^H_{10})^{-\alpha},$$

By (36), we have that $\Lambda\,(\bar{k}_1 - \bar{p}^H_{10})^{-\alpha} = \bar{k}_2 - \bar{p}^c_{11|10}$. Replacing it and combining both derivatives:

$$\frac{\partial S^*}{\partial\bar{p}^H_{10}} = \frac{1}{\bar{P}^H} - \frac{\bar{p}^c_{11|10}}{\bar{P}^H\bar{P}^c} = \frac{1}{\bar{P}^H}\left(1 - \frac{\bar{p}^c_{11|10}}{\bar{P}^c}\right) = \frac{1-\theta_{c,10}}{\bar{P}^H} > 0, \tag{40}$$

with $\theta_{c,10} \equiv \bar{p}^c_{11|10}/\bar{P}^c$, strictly positive because $\theta_{c,10} < 1$ under Assumption A.4.

S^* is strictly increasing in the college threshold function. Using the integral form of (38), in which $\bar{p}^c_{11|00}(\cdot)$ appears explicitly as the integrand. Since S^* is an affine functional of $\bar{p}^c_{11|00}(\cdot)$ with strictly positive coefficient $1/(\bar{P}^H\bar{P}^c)$, the functional derivative at any $p^H_0 \in (\bar{p}^H_{10}, \bar{P}^H)$ is

$$\frac{\delta S^*}{\delta\,\bar{p}^c_{11|00}(p^H_0)} = \frac{1}{\bar{P}^H\bar{P}^c} > 0.$$

So any parameter change that raises $\bar{p}^c_{11|00}(p^H)$ within $[\bar{p}^H_{10}, \bar{P}^H]$, and lowers it nowhere, strictly raises S^*.

A.5 Necessary Conditions

The share of individuals completing high school might move as a result of changes in two channels: the indirect labor market channels, which moves students from $(0,0)$ to $(1,0)$ and increases S^* by increasing θ_H , and the direct education effect via transitions from $(0,0)$ to $(1,1)$, which operates through the second term. In this section, I derive the necessary conditions for these transitions to take place.

A.5.1 Condition 1: a non-empty pool at $(0,0)$, $\bar{p}^H_{10} < \bar{P}^H$

For either channel to be active, the path $(0,0)$ need to attract at least some students pre-policy. A student is at $(0,0)$ if and only if $p^H > \bar{p}^H_{10}$ *and* $p^c > \bar{p}^c_{11|00}(p^H)$. Conditioning on p^H and using independence of the two draws,

$$\begin{aligned}\Pr\big((s,d)^* = (0,0)\big) &= \int_{\bar{p}^H_{10}}^{\bar{P}^H} \Pr\big(p^c > \bar{p}^c_{11|00}(p^H)\,\big|\,p^H\big)\, f(p^H)\, dp^H = \int_{\bar{p}^H_{10}}^{\bar{P}^H} \frac{\bar{P}^c - \bar{p}^c_{11|00}(p^H)}{\bar{P}^c} \cdot \frac{1}{\bar{P}^H}\, dp^H \\ &= \frac{1}{\bar{P}^H \bar{P}^c} \int_{\bar{p}^H_{10}}^{\bar{P}^H} \big[\bar{P}^c - \bar{p}^c_{11|00}(p^H)\big]\, dp^H,\end{aligned}$$

where the middle step uses $\Pr(p^c > \bar{p}^c_{11|00}(p^H)) = 1 - \bar{p}^c_{11|00}(p^H)/\bar{P}^c$, which is valid because Assumption A.4 places $\bar{p}^c_{11|00}(p^H)$ inside $(0, \bar{P}^c)$.

This condition implies that $\bar{p}^H_{10} < \bar{P}^H$. Assume $\bar{p}^H_{10} \geq \bar{P}^H$. Then every student draws $p^H \leq \bar{P}^H \leq \bar{p}^H_{10}$, so every student graduates and $S^* = 1$. Hence no parameter change can raise S^*, and $\bar{p}^H_{10} < \bar{P}^H$ is necessary. Recall the formula for $\bar{p}^H_{10}$:

$$\bar{p}^H_{10} = \bar{k}_1 - (\bar{k}_1 + y_1^{NH}) \left(\frac{\bar{k}_2 + y_2^{NH}}{\bar{k}_2 + y_2^{HS}}\right)^{\beta_2} \left(\frac{\bar{k} + y^{NH}}{\bar{k} + y^{HS}}\right)^{\beta_3}.$$

As $y^{HS} \to \infty$ both ratios tend to 0 and $\bar{p}^H_{10} \to \bar{k}_1$. If $\bar{k}_1 \geq \bar{P}^H$, then $\theta_H \to 1$ and the pool at $(0,0)$ disappears. Condition 1 therefore places an *upper* bound on the high-school wage premium: it must not be so large that all students graduate even in the complete absence of any college option.

A.5.2 Condition 2: a reachable pool

Condition 1 guarantees that students sit at $(0,0)$. However, it does not guarantee that either transitions can take place. That is, that students sitting at $(0,0)$ can move to either $(1,0)$ or $(1,1)$. Below, I derive the necessary conditions for each transition to take place.

Necessary conditions for the indirect labor market channel. For $(0,0)$ to $(1,0)$ transitions to take place, it needs to be true that $p^H < \bar{p}^H_{10}$ is attainable for at least some students. That implies that $\bar{p}^H_{10}$ must be positive, since $p^H > 0$.

From:

$$\bar{p}^H_{10} = \bar{k}_1 - (\bar{k}_1 + y_1^{NH}) \left(\frac{\bar{k}_2 + y_2^{NH}}{\bar{k}_2 + y_2^{HS}}\right)^{\beta_2} \left(\frac{\bar{k} + y^{NH}}{\bar{k} + y^{HS}}\right)^{\beta_3}.$$

$\bar{p}^H_{10} > 0$ if and only if $(\bar{k}_1 + y_1^{NH}) \left(\frac{\bar{k}_2 + y_2^{NH}}{\bar{k}_2 + y_2^{HS}}\right)^{\beta_2} \left(\frac{\bar{k} + y^{NH}}{\bar{k} + y^{HS}}\right)^{\beta_3} < \bar{k}_1$. Dividing by $(\bar{k}_1 + y_1^{NH}) > 0$ and substituting $\bar{k}_1 = 3\bar{k}$, $y_1^{NH} = 3y^{NH}$ on the right,

$$\frac{\bar{k}_1}{\bar{k}_1 + y_1^{NH}} = \frac{3\bar{k}}{3(\bar{k} + y^{NH})} = \frac{\bar{k}}{\bar{k} + y^{NH}},$$

so the exact necessary and sufficient condition is

$$\bar{p}^H_{10} > 0 \iff \left(\frac{\bar{k}_2 + y_2^{NH}}{\bar{k}_2 + y_2^{HS}}\right)^{\beta_2} \left(\frac{\bar{k} + y^{NH}}{\bar{k} + y^{HS}}\right)^{\beta_3} < \frac{\bar{k}}{\bar{k} + y^{NH}}. \tag{41}$$

If $y^{HS} \leq y^{NH}$ then $y_2^{HS} \leq y_2^{NH}$, so both ratios are ≥ 1. Since $y^{NH} > 0$, $\bar{k}/(\bar{k} + y^{NH}) < 1$. Thus, $y^{HS} \leq y^{NH}$ implies that $\bar{p}^H_{10} \leq 0$. Contrapositively, $\bar{p}^H_{10} > 0 \Rightarrow y^{HS} > y^{NH}$.

Necessary conditions for the direct education channel. For $(0,0)$ to $(1,1)$ transitions to take place, it needs to be true that $p^c < \bar{p}^c_{11|00}$ is attainable for at least some students. That implies that $\bar{p}^c_{11|00}$ is positive, since $p^c > 0$. By (35), if $\bar{p}^c_{11|10} \leq 0$, then $\bar{p}^c_{11|00} < 0$. Thus, $\bar{p}^c_{11|10} > 0$ implies $\bar{p}^c_{11|00} > 0$. I can then study when $\bar{p}^c_{11|10} > 0$ to determine the necessary conditions for $\bar{p}^c_{11|00} > 0$.

Using $\bar{k}_2 = 4\bar{k}$ and $\bar{k}_2 + y_2^{HS} = 4(\bar{k} + y^{HS})$:

$$\bar{p}^c_{11|10} = 4\left[\bar{k} - (\bar{k} + y^{HS})\left(\frac{\bar{k} + y^{HS}}{\bar{k} + y^C}\right)^{\beta_3/\beta_2}\right].$$

Note that at $y^C = y^{HS}$ the ratio equals 1 and $\bar{p}^c_{11|10} = 4[\bar{k} - (\bar{k} + y^{HS})] = -4y^{HS} < 0$. As $y^C \to \infty$ the ratio tends to 0 and $\bar{p}^c_{11|10} \to 4\bar{k} > 0$. Since $\bar{p}^c_{11|10}$ is increasing in y^C, there is a unique threshold $\bar{y}^C$ where $\bar{p}^c_{11|10} = 0$, and for which any $y^C > \bar{y}^C$ implies that $\bar{p}^c_{11|10} > 0$. Then $\bar{p}^c_{11|10} > 0 \iff y^C > \bar{y}^C$. Setting $\bar{p}^c_{11|10} = 0$ and dividing by $4(\bar{k} + y^{HS})$:

$$\frac{\bar{k}}{\bar{k} + y^{HS}} = \left(\frac{\bar{k} + y^{HS}}{\bar{k} + \bar{y}^C}\right)^{\beta_3/\beta_2}.$$

Raising both sides to the power $\beta_2/\beta_3 > 0$ and multiplying by $(\bar{k} + y^{HS})$:

$$\bar{y}^C = (\bar{k} + y^{HS})\left(\frac{\bar{k} + y^{HS}}{\bar{k}}\right)^{\beta_2/\beta_3} - \bar{k}. \tag{42}$$

Since $y^{HS} > 0$, it follows that $\big((\bar{k} + y^{HS})/\bar{k}\big)^{\beta_2/\beta_3} > 1$, and thus $\bar{y}^C > y^{HS}$ strictly. Therefore, a necessary condition for $\bar{p}^c_{11|10} > 0$, and consequently, $\bar{p}^c_{11|00} > 0$, is that $y^C > \bar{y}^C > y^{HS}$.

A.6 Comparative Statics

A.6.1 The chain rule

To analyze how the share of students completing high school changes with policy-driven changes in parameters, I differentiate the closed form expression (39) with respect to key parameters. Recall $\mathcal{S}$ denotes the braced expression, and by construction $\mathcal{S} = \int_{\bar{p}^H_{10}}^{\bar{P}^H} \bar{p}^c_{11|00}(p^H)\, dp^H$. For a given parameter x,

$$\frac{\partial S^*}{\partial x} = \frac{1}{\bar{P}^H}\frac{\partial \bar{p}^H_{10}}{\partial x} + \frac{1}{\bar{P}^H \bar{P}^c}\frac{d\mathcal{S}}{dx}, \qquad \frac{d\mathcal{S}}{dx} = \frac{\partial \mathcal{S}}{\partial \bar{p}^H_{10}}\frac{\partial \bar{p}^H_{10}}{\partial x} + \frac{\partial \mathcal{S}}{\partial x}\bigg|_{\bar{p}^H_{10}\text{ fixed}},$$

where $\partial\mathcal{S}/\partial\bar{p}^H_{10} = -\bar{p}^c_{11|10}$. Collecting the terms in $\partial\bar{p}^H_{10}/\partial x$ and using $1/\bar{P}^H - \bar{p}^c_{11|10}/(\bar{P}^H\bar{P}^c) = (1 - \theta_{c,10})/\bar{P}^H$:

$$\frac{\partial S^*}{\partial x} = \underbrace{\frac{1-\theta_{c,10}}{\bar{P}^H}\cdot\frac{\partial \bar{p}^H_{10}}{\partial x}}_{(0,0)\text{ to }(1,0)} + \underbrace{\frac{1}{\bar{P}^H\bar{P}^c}\int_{\bar{p}^H_{10}}^{\bar{P}^H}\frac{\partial \bar{p}^c_{11|00}(p^H)}{\partial x}\, dp^H}_{(0,0)\text{ to }(1,1)}. \tag{43}$$

A.6.2 Signing the college-threshold term

Define $\Delta(p^c, x; p^H) \equiv V(1,1) - V(0,0)$:

$$V(1,1) - V(0,0) = \ln\left(\frac{\bar{k}_1 - p^H}{\bar{k}_1 + y_1^{NH}}\right) + \beta_2 \ln\left(\frac{(\bar{k}_2 + y_2^{HS})^{1-\pi_c}(\bar{k}_2 - p^c)^{\pi_c}}{\bar{k}_2 + y_2^{NH}}\right) + \beta_3 \ln\left(\frac{(\bar{k} + y^{HS})^{1-\pi_c}(\bar{k} + y^C)^{\pi_c}}{\bar{k} + y^{NH}}\right), \tag{44}$$

So that $\Delta = 0$ at $p^c = \bar{p}^c_{11|00}(p^H)$. The only p^c-dependent term is $\pi_c\beta_2\ln(\bar{k}_2 - p^c)$, so

$$\frac{\partial\Delta}{\partial p^c} = -\frac{\pi_c\beta_2}{\bar{k}_2 - p^c} < 0,$$

which is strictly negative because $p^c < \bar{k}_2$. The implicit function theorem therefore gives

$$\frac{\partial \bar{p}^c_{11|00}(p^H)}{\partial x} = -\frac{\partial \Delta/\partial x}{\partial \Delta/\partial p^c} = \frac{\bar{k}_2 - \bar{p}^c_{11|00}(p^H)}{\pi_c \beta_2} \cdot \frac{\partial \Delta}{\partial x}, \tag{45}$$

Since $\frac{\bar{k}_2 - \bar{p}^c_{11|00}(p^H)}{\pi_c \beta_2} > 0$, then $\operatorname{sign}\big(\partial \bar{p}^c_{11|00}/\partial x\big) = \operatorname{sign}\big(\partial \Delta/\partial x\big)$, and it suffices to sign $\partial \Delta/\partial x$.

A.6.3 Effect of p^c via the support $\bar{P}^c$

$\bar{P}^c$ enters S^* only through the explicit factor $1/\bar{P}^c$ in (38), and

$$\frac{\partial S^*}{\partial \bar{P}^c} = -\frac{1}{\bar{P}^H\, \bar{P}^{c2}} \int_{\bar{p}^H_{10}}^{\bar{P}^H} \bar{p}^c_{11|00}(p^H)\, dp^H. \tag{46}$$

Under the necessary condition that $\bar{p}^c_{11|10} > 0$, $\bar{p}^c_{11|00}(\bar{p}^H_{10}) = \bar{p}^c_{11|10} > 0$ by (33) and, by continuity, $\bar{p}^c_{11|00} > 0$ on a set of positive measure, so the integral is strictly positive and (46) is strictly negative. On the other hand, if the necessary condition is not met and If $\bar{p}^c_{11|10} \le 0$, then by (35) $\bar{p}^c_{11|00}(p^H) \le 0$ throughout the range, and $\Pr(p^c \le \bar{p}^c_{11|00}(p^H)) = 0$ for every student. The integral is zero, and $\partial S^*/\partial \bar{P}^c = 0$. Hence

$$\frac{\partial S^*}{\partial \bar{P}^c} < 0 \quad \text{if and only if} \quad \bar{p}^c_{11|10} > 0, \tag{47}$$

so lowering college costs unambiguously raises S^* exactly when the college channel is non-empty.

A.6.4 Effect of π_c

Changes in S^* brought about by changes in π_c run only through the educational opportunities channel, so $\partial \bar{p}^H_{10}/\partial \pi_c = 0$. Then, through the education opportunities channel, we have that:

$$\frac{\partial \Delta}{\partial \pi_c} = \beta_2 \ln\!\left(\frac{\bar{k}_2 - p^c}{\bar{k}_2 + y_2^{HS}}\right) + \beta_3 \ln\!\left(\frac{\bar{k} + y^C}{\bar{k} + y^{HS}}\right), \tag{48}$$

The first term of (48) is strictly negative for *every* student and every admissible p^c, because $y_2^{HS} > 0$ and $p^c > 0$ give $\bar{k}_2 - p^c < \bar{k}_2 < \bar{k}_2 + y_2^{HS}$. This is the period-2 cost of raising the admission probability, since a larger π_c makes the student more likely to be paying p^c instead of earning y_2^{HS}. The second term is the period-3 benefit, which is positive when $y^C > y^{HS}$. At an *arbitrary* p^c, the sign of (48) is sign is ambiguous even when $y^C > y^{HS}$, since which term will dominate depends on the size of p^c.

By (45), $\partial \bar{p}^c_{11|00}(p^H)/\partial \pi_c$ is $\partial \Delta/\partial \pi_c$ evaluated *at* $p^c = \bar{p}^c_{11|00}(p^H)$, which is the point of indifference (threshold) and the only point that moves when π_c changes. By (35), $\bar{p}^c_{11|00}(p^H) < \bar{p}^c_{11|10}$ for $p^H > \bar{p}^H_{10}$, so $\bar{k}_2 - \bar{p}^c_{11|00}(p^H) > \bar{k}_2 - \bar{p}^c_{11|10}$, and by (30) the latter equals $(\bar{k}_2 + y_2^{HS})\big((\bar{k} + y^{HS})/(\bar{k} + y^C)\big)^{\beta_3/\beta_2}$. Dividing by $(\bar{k}_2 + y_2^{HS}) > 0$, taking logarithms and multiplying by $\beta_2 > 0$,

$$\beta_2 \ln\left(\frac{\bar{k}_2 - \bar{p}^c_{11|00}(p^H)}{\bar{k}_2 + y_2^{HS}}\right) > -\beta_3 \ln\left(\frac{\bar{k} + y^C}{\bar{k} + y^{HS}}\right),$$

i.e. $\partial\Delta/\partial\pi_c > 0$ at the threshold.

Under the necessary condition that $\bar{p}^c_{11|10} > 0$, $\partial\bar{p}^c_{11|00}(p^H)/\partial\pi_c > 0$ for every $p^H \in (\bar{p}^H_{10}, \bar{P}^H)$. If $\bar{p}^c_{11|10} \leq 0$ then $\bar{p}^c_{11|00}(p^H) \leq 0$ throughout, and the college channel is identically zero for every π_c. Hence:

$$\frac{\partial S^*}{\partial \pi_c} = \frac{1}{\bar{P}^H \bar{P}^c} \int_{\bar{p}^H_{10}}^{\bar{P}^H} \frac{\partial \bar{p}^c_{11|00}(p^H)}{\partial \pi_c}\, dp^H > 0 \quad \text{if and only if} \quad \bar{p}^c_{11|10} > 0. \tag{49}$$

A.6.5 Effect of y^{HS}

Changes in high-school level wages operate through both the indirect labor market and the direct educational channel. Let:

$\bar{p}^H_{10} = \bar{k}_1 - (\bar{k}_1 + y_1^{NH})\Pi$ with

$$\ln \Pi = \beta_2 \ln\left(\frac{\bar{k}_2 + y_2^{NH}}{\bar{k}_2 + y_2^{HS}}\right) + \beta_3 \ln\left(\frac{\bar{k} + y^{NH}}{\bar{k} + y^{HS}}\right),$$

so $\partial \ln \Pi/\partial y^{HS} = -(\beta_2 + \beta_3)/(\bar{k} + y^{HS})$

Using $\bar{k}_1 + y_1^{NH} = 3(\bar{k} + y^{NH})$ throughout,

$$\frac{\partial \bar{p}^H_{10}}{\partial y^{HS}} = \frac{3(\bar{k} + y^{NH})(\beta_2 + \beta_3)\,\Pi}{\bar{k} + y^{HS}} \;>\; 0, \tag{50}$$

On the direct educational channel, we have that:

$$\frac{\partial \Delta}{\partial y^{HS}} = \frac{(1 - \pi_c)(\beta_2 + \beta_3)}{\bar{k} + y^{HS}} \;>\; 0, \tag{51}$$

Thus, higher high-school wages raise the high school threshold, pulling students from $(0, 0)$ into $(1, 0)$, and the college threshold rises too, because y^{HS} is earned in the non-admission state on the college path. Both terms of (43) are strictly positive, so

$$\frac{\partial S^*}{\partial y^{HS}} > 0 \tag{52}$$

A.6.6 Effect of y^{NH}

Similarly, changes in primary-educated wages operate through both channels. Let: $\bar{p}^H_{10} = \bar{k}_1 - (\bar{k}_1 + y^{NH}_1)\Pi$ with

$$\ln \Pi = \beta_2 \ln\left(\frac{\bar{k}_2 + y^{NH}_2}{\bar{k}_2 + y^{HS}_2}\right) + \beta_3 \ln\left(\frac{\bar{k} + y^{NH}}{\bar{k} + y^{HS}}\right),$$

so $\partial \ln \Pi / \partial y^{NH} = +(\beta_2 + \beta_3)/(\bar{k} + y^{NH})$. Using $\bar{k}_1 + y^{NH}_1 = 3(\bar{k} + y^{NH})$ throughout,

$$\frac{\partial \bar{p}^H_{10}}{\partial y^{NH}} = -3\,\Pi\,(1 + \beta_2 + \beta_3) \;<\; 0, \tag{53}$$

On the direct educational channel, we have:

$$\frac{\partial \Delta}{\partial y^{NH}} = -\frac{1 + \beta_2 + \beta_3}{\bar{k} + y^{NH}} \;<\; 0, \tag{54}$$

Both termsof (43) are strictly negative, so

$$\frac{\partial S^*}{\partial y^{NH}} < 0 \tag{55}$$

A.6.7 Effect of y^C

College earnings operate solely through the direct education channel. We have:

$$\frac{\partial \Delta}{\partial y^{C}} = \frac{\beta_3 \pi_c}{\bar{k} + y^{C}} \;>\; 0, \tag{56}$$

And thus:
$$\frac{\partial \bar{p}^c_{11|00}(p^H)}{\partial y^C} = \frac{\bar{k}_2 - \bar{p}^c_{11|00}(p^H)}{\pi_c \beta_2} \cdot \frac{\beta_3 \pi_c}{\bar{k} + y^C} = \frac{\beta_3\left(\bar{k}_2 - \bar{p}^c_{11|00}(p^H)\right)}{\beta_2(\bar{k} + y^C)} \;>\; 0,$$

the factor π_c cancelling. Hence

$$\frac{\partial S^*}{\partial y^C} = \frac{\beta_3}{\beta_2\,\bar{P}^H\,\bar{P}^c\,(\bar{k} + y^C)} \int_{\bar{p}^H_{10}}^{\bar{P}^H} \left(\bar{k}_2 - \bar{p}^c_{11|00}(p^H)\right) dp^H > 0 \quad \text{if and only if} \quad \bar{p}^c_{11|10} > 0. \tag{57}$$

Under the necessary condition that $\bar{p}^c_{11|10} > 0$, an increase in college-level wages should increase the share of students graduating high school.